1st Ishaan

2nd Connor

3rd Ayla

4th Sophi

5th Julia

6th Spencer

7th Ravi

8th Tomiwa

9th Shannon

PATHOGENESIS/PATHOPHYSIOLOGY (normal → disease state)

1. Describe the structural and functional anatomy of the organs of the reproductive system and external genitalia of individuals assigned female at birth, including their surface projections and relationships to other pelvic structures. ANAT **Connor**
2. Discuss the microscopic changes seen in the ovary and uterus during the ovarian cycle, their correlations to normal hormonal fluctuations and be able to identify physiologically relevant cells/tissues by microscopy. ANAT **Sophie**
3. Describe the mechanistic basis for the advantages of transcription-mediated amplification (TMA) over the polymerase chain reaction (PCR). BCM **Ravi**
4. Describe the function of the estrogen and progesterone receptors and diagram their signaling pathways. BCM **Ishaan**
5. Describe gonadotropin control of ovarian function and outline the phases of the ovarian cycle with the corresponding hormonal patterns. PHYSIO **Ishaan**
6. Describe the principal physiologic actions of estrogen and progesterone on their target organs. PHYSIO **Ishaan**
7. Describe the trigger for and physiologic process of normal menarche. PHYSIO **Shannon**
8. Review the microbiology, pathogenesis, and clinical manifestations of herpes simplex virus and human papilloma virus. INIM JP
9. Describe the microbiologic characteristics of the bacteria associated with urethritis and cervicitis (focus on Neisseria gonorrhea, Chlamydia trachomatis, Mycoplasma genitalium). INIM JP
10. Explain the pathogenesis, clinical manifestations and complications of infections affecting the vulva, vagina, cervix, and fallopian tubes, including pelvic inflammatory disease. INIM & PATH JP
11. Explain the pathophysiologic basis of polyendocrine metabolic ovarian syndrome (PMOS), including the contributions of hyperandrogenism, disrupted gonadotropin secretion, and insulin resistance to its clinical presentation. PATH **Sophie**
12. Compare and contrast common malformations of the Müllerian system and lower reproductive tract, and identify those that may present as primary amenorrhea. PATH-**Tomiwa**

EPIDEMIOLOGY (incidence, prevalence, risk factors)
CLINICAL PRESENTATION (signs, symptoms, history, exam)

1. Describe and apply the components of the Tanner stages. CS- **Connor**
2. Compare and contrast the history and physical exam elements in primary versus secondary amenorrhea. CS - **Ayla P**

DIAGNOSIS (differential diagnosis, diagnostic tests and tools, clinical reasoning)

1. Categorize the causes of primary amenorrhea by the following etiologies: hypothalamic/pituitary disease, ovarian disease, uterine/outflow tract disorders, receptor/enzyme deficiencies. CM - **Ayla P**
2. Compare and contrast primary and secondary amenorrhea and the general diagnostic approach to evaluation. CM - **Ayla P**
3. Describe the screening recommendations and available diagnostic tests for chlamydia, gonorrhea, trichomonas, HSV and syphilis. INIM - **Spencer**

TREATMENT AND OUTCOMES

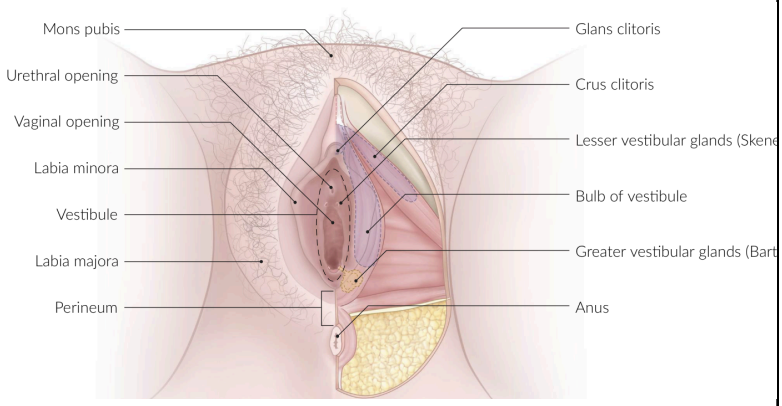
1. Describe and identify the morphology of endometrial hyperplasia and endometrial carcinoma; explain the pathogenesis of these conditions, including why a patient with PMOS is at increased risk for their development. PATH-**Tomiwa**
2. Compare their efficacy, complications, and contraindications for contraceptive options. CM and PHARM **Ravi**
3. Describe the treatment strategies for the common STis (empiric vs targeted therapy and treatment of contacts). CM - **Spencer**

4. Describe the preferred treatment regimens, their mechanism of action, and adverse effects for Neisseria gonorrhea and Chlamydia trachomatis as recommended by the CDC. PHARM **Shannon**
5. Discuss the potential long-term implications of HPV infection and the rationale for HPV vaccination. HSS (Please review general HPV information and HPV vaccine guidelines on the CDC website, with specific attention to cancers linked to HPV and vaccine recommendations. **Shannon**
   Available at: https://www.cdc.gov/hpv/index.html)

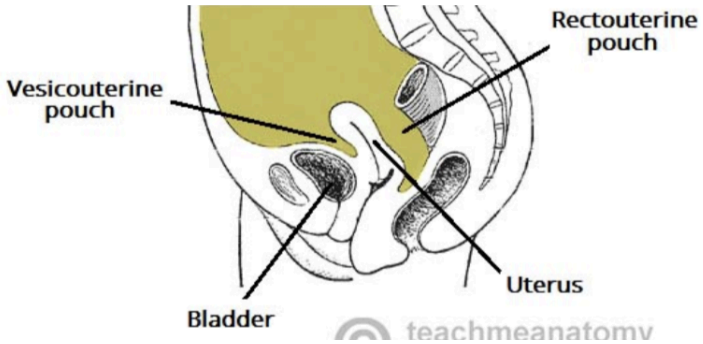
Describe the structural and functional anatomy of the organs of the reproductive system and external genitalia of individuals assigned female at birth, including their surface projections and relationships to other pelvic structures. ANAT

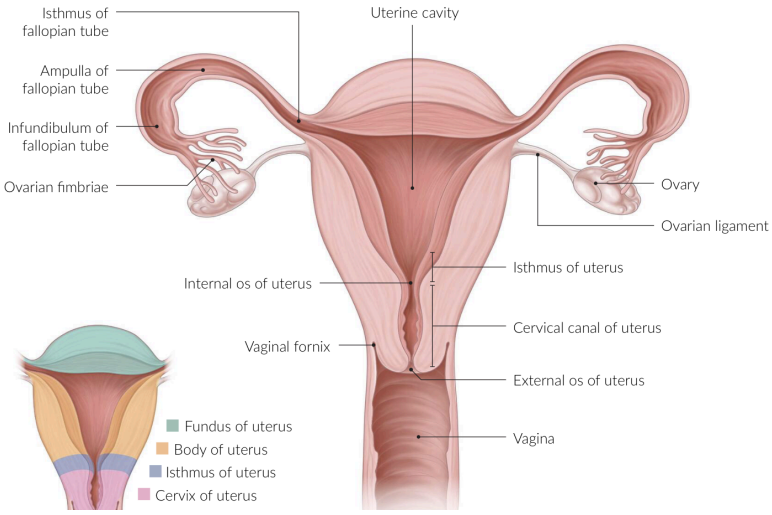
External Genitalia → The Vulva

- **Labia majora**: lateral folds, pubic hair
- **Labia minora**: folds surrounding vestibule
- **Vulva vestibule**: contains vaginal and urethral orifices
- **Greater vestibular glands (Bartholin)**: secrete alkaline fluid for lubrication
- **Lesser vest glands (Skene)**: function unknown, homologous to male prostate gland
- **Bulbs of vestibule**: erectile tissue that flanks the vaginal orifice
- **Clitoris**: erectile tissue composed of corpora cavernosa



Internal Genitalia



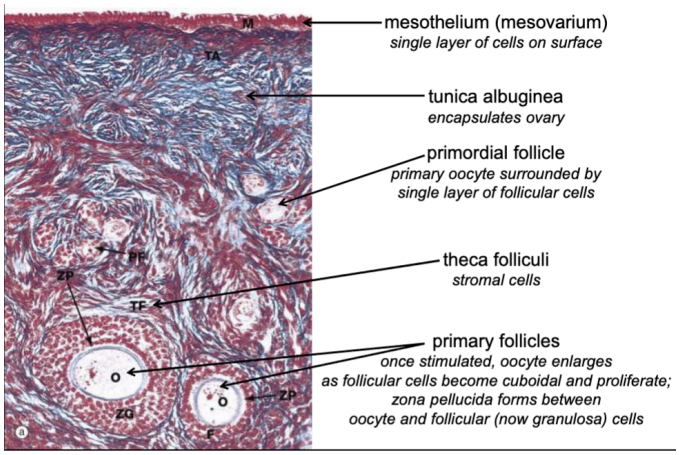


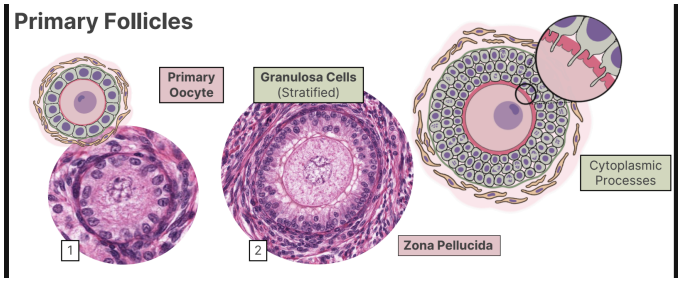
Vagina ● Extraperitoneal within the pelvis; extends from uterus → vaginal vestibule ● Passageway for menstruation and childbirth ● Barrier against infections ○ pH 4-4.5 ● Hymen: membrane the partially covers vaginal opening	**Cervix** ● Low, narrow end of uterus ● Forms a canal between uterus and vagina ● Secretes alkaline fluid to promote sperm flow ● Ectocervix → outer part of the cervix (facing vagina) ● Endocervix → inner portion of cervix (facing uterus) ● **Anteverted**: 90° between cervix and vagina ● **Anteflexed**: 100-120° between cervix and uterine body
Uterus ● Site of implantation and menstruation ● Protects and supports fetus ● Uterine horns: insertion point of fallopian tubes ● Posterior to bladder ● Anterior to rectum	**Fallopian Tubes** ● suspended within broad ligament ● Typical site of fertilization ● Fimbria sweep ovum from ovary at ovulation → infundibulum → ampulla → isthmus → uterine horn
Ovary ● Site of oogenesis and sex hormone synthesis ● Suspended in broad ligament ● Anchored by ovarian ligament	**Other** ● Uterovesical pouch: between bladder and uterus ● Rectouterine pouch: between rectum and uterus

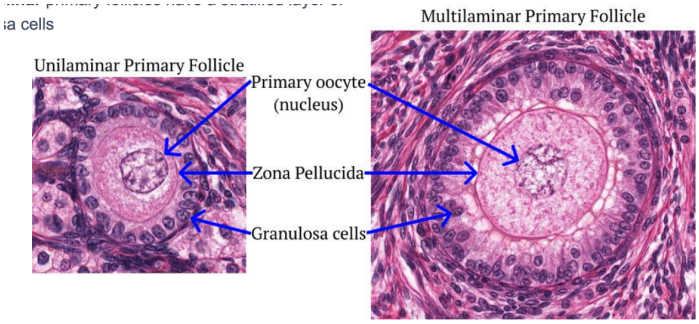
1. Discuss the microscopic changes seen in the ovary and uterus during the ovarian cycle, their correlations to normal hormonal fluctuations and be able to identify physiologically relevant cells/tissues by microscopy. ANAT **Sophie**

Summary (still look at the histo tho):

- The **follicular phase** is from day 0-14 and is mainly driven by **estrogen** and **FSH**
  - In the ovary:
    - the follicle will mature from a **primordial follicle** to **mature graafian follicle**
  - In the uterus:
    - the endometrium will first go through menses, and then be in the **proliferative phase** where it histologically looks uniform
- On day 14, the **LH** surge will cause the rupture of the mature follicle, expelling the oocyte into the fallopian tubes, leaving behind the follicle
- The **luteal phase** is from day 14-28 governed by **progesterone**
  - In the ovary:
    - The left behind follicle will first become the **corpus luteum** which secretes **progesterone** and **estrogen** which maintains the endometrium.
    - After 10-14 days with no fertilization occurring, the corpus luteum will degrade into the fibrous **corpus albicans**, which can last for many months
  - In the uterus:
    - The endometrium will be in the **secretory phase,** where it histologically looks weird

	Follicular Phase (day 0-14)	**Luteal Phase (day 14-28)**
Ovary	**PRIMARY FOLLICLES**  	**DAY 14: theca externa** will contract and expel and rupture the follicle, expelling the oocyte from the ovary driven by a surge in **LH** **CORPUS LUTEUM (Day 14-24)**

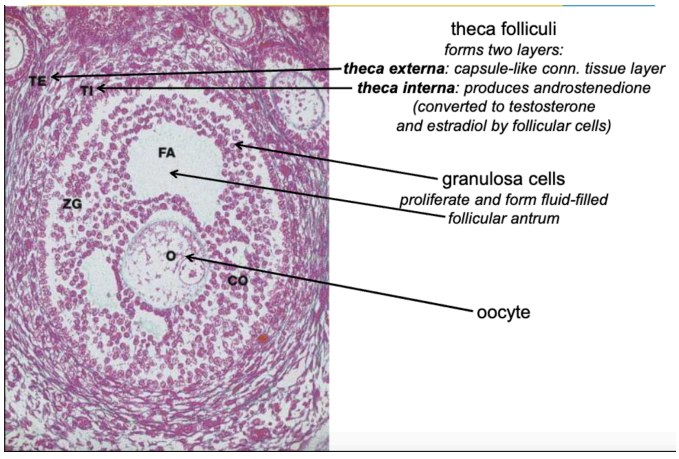


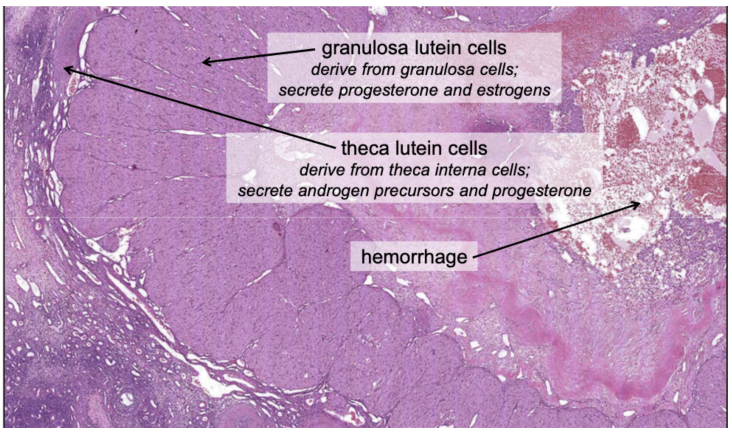


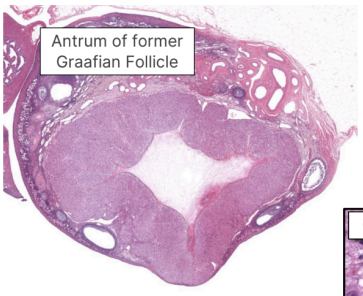
Primordial follicles (***gestation until cycle it is used in***) surrounded by single layer of follicular cell, cells are arrested in **prophase I**

Primary follicles (***Day 0-7?)***)oocyte is selected, still in prophase I, with stimulation from **FSH,** follicular cells become multi-layered-> nomenclature switch follicular cells now called **granulosa cells.**
->**Zona pellucida** forms between oocyte and granulosa cells, glycoprotein rich matrix

SECONDARY FOLLICLES

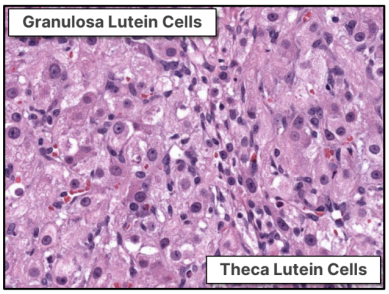






C

LH stimulates the **ruptured Graafian foll** luteum in the ovary **after ovulation**, a hig gland. **Luteal hormones prepare uter** days 15-28 of the ovarian and menstrual **hCG** hormone prevents it



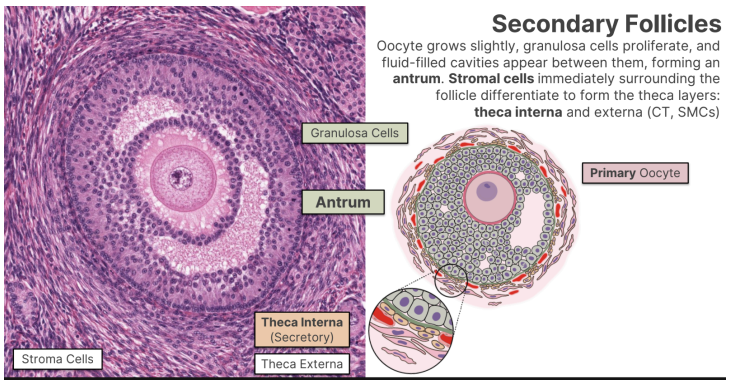
Granulosa cells become **Granulosa Lutein cells** (synthesize estrogen and **progesterone**)

Theca interna cells become **Theca Lutein cells** (synthesize androgens and **progesterone**)

Oocyte was expelled from the ovary, leaving granulosa cells behind in the ovary, which become the **granulosa lutein,** which secrete **progesterone, inhibin A,** and **estrogen** which act on endometrial lining (this is maintained by **B-HCG** if pregnancy is established).

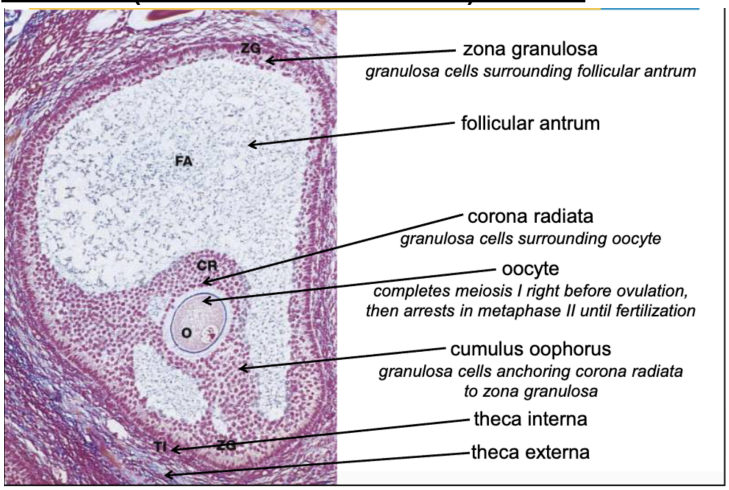
Similarly the theca interna cells become **theca lutein cells,** and secrete **progesterone**

CORPUS ALBICANS (Day 24 through multiple cycles later)



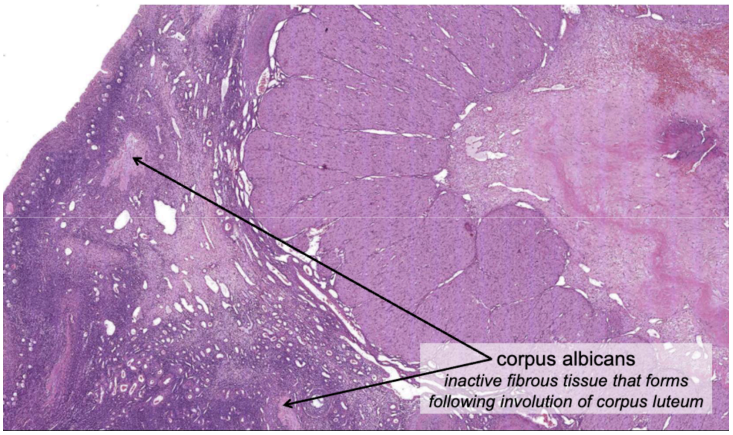
Secondary follicles (***day 7?-13?***) characterized by a fluid filled antrum, have both a…
Theca interna: light staining, with **LH** produce **androgens,** quickly converted to **estrogen**
Theca externa: darker cells, contractile

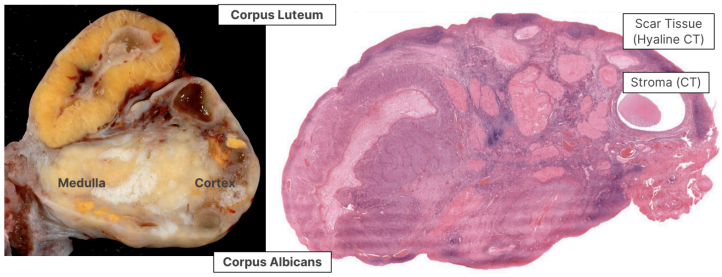
MATURE (GRAAFIAN FOLLICLES) *DAY 14*



The **mature graafian follicle** is what emerges from the ovary during ovulation, >10mm in diameter
****only 1 develops and emerges each cycle**
->at this point follicle proceeds from prophase I to metaphase II (leaves behind polar body)
->**corona radiata** immediately surrounds oocytes
->**zona granulosa** is the border surrounding the liquid follicular antrum, border of the follicle
->**cumulus oophorus** connects corona radiata to zona granulosa

DAY 14: **theca externa** will contract rupture the follicle, expelling the oocyte from the ovary driven by a surge in **LH**

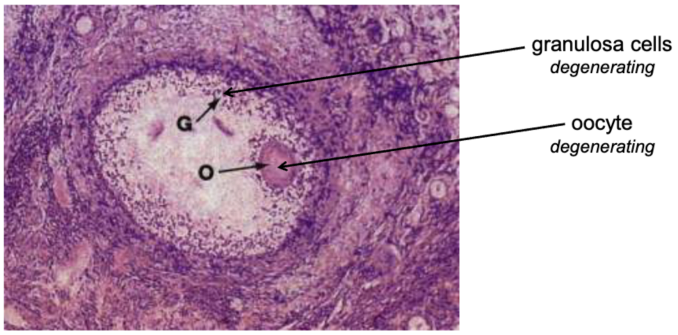




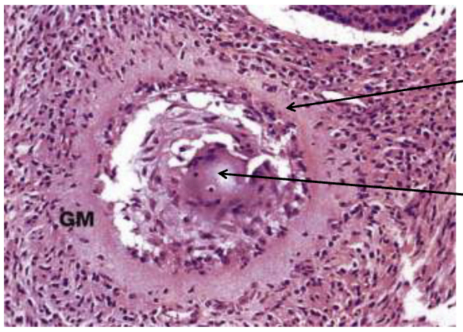
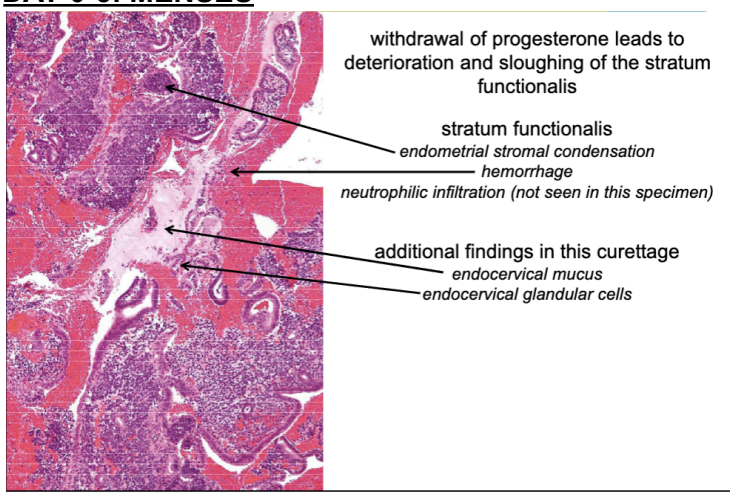
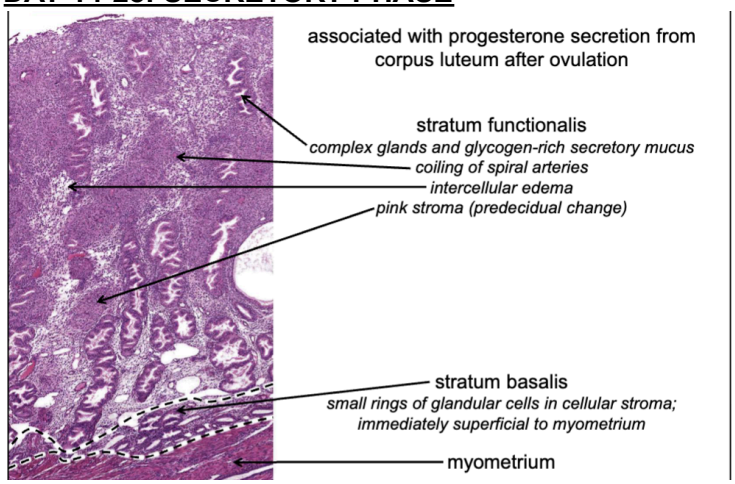
In the absence of fertilization (lack of BHCG), corpus luteum will be degenerate and become fibrous tissue "*white body*" or **corpus albicans**

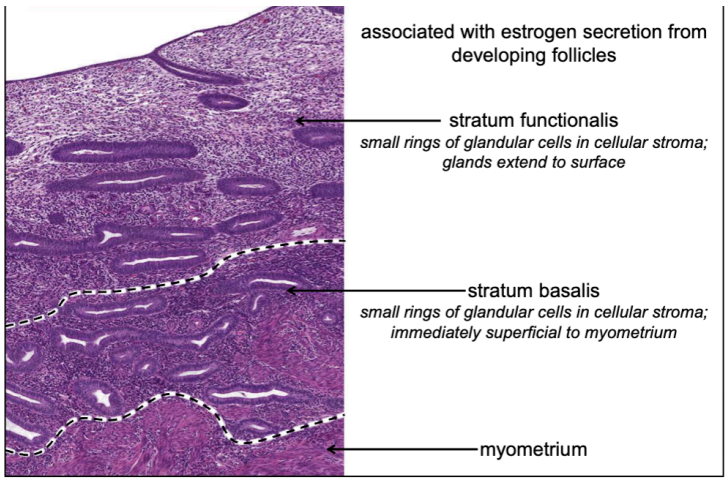
ATRESIA: (if the follicle is NOT chosen to ovulate)

Simultaneous but separate process that is still happening in ovary to 19-29 follicles
Early changes:



Late changes:

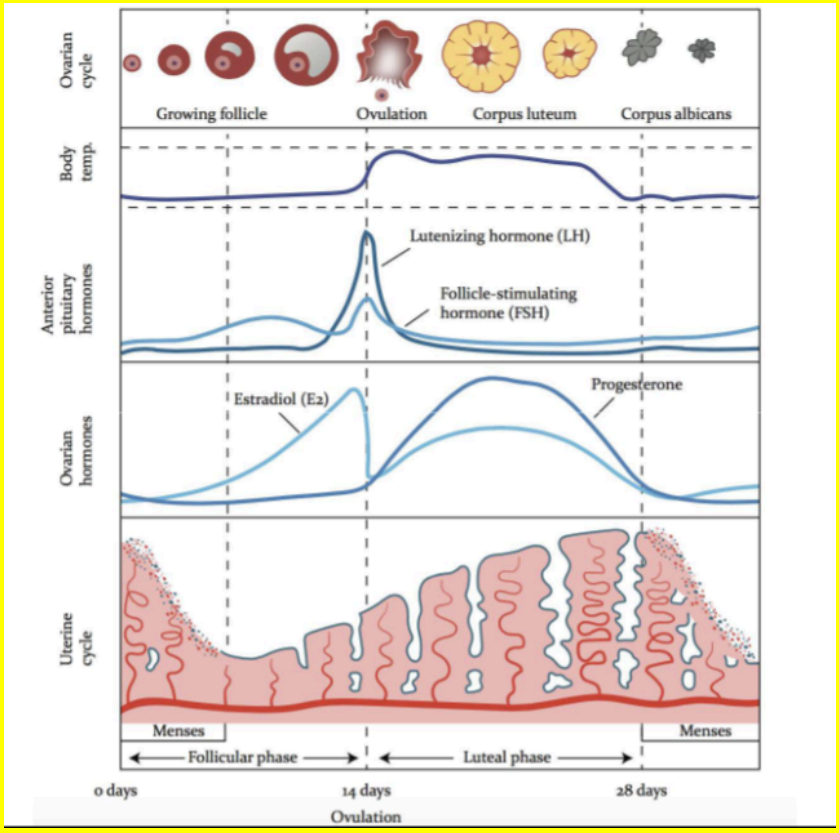
		
Uterus	**DAY 0-5: MENSES**  Driven by the withdrawal of **progesterone** The **stratum functionalis** shows deterioration and sloughing. The stroma will condense, forming dense balls of nuclei. THere will also be neutrophilic infiltration Also **hemorrhage** **DAY 6-14: PROLIFERATIVE PHASE**	**DAY 14-28: SECRETORY PHASE**  Driven by **progesterone** secretion from the **corpus luteum** In this phase, the **stratum functionalis** does not appear similar to the **stratum basalis** **Stratum functionalis** has complex, corkscrew-shaped glands with paler cytoplasm, due to high glycogen **Stratum basalis** looks the same as it did in proliferative phase: small rings of glandular cells in cellular stroma.

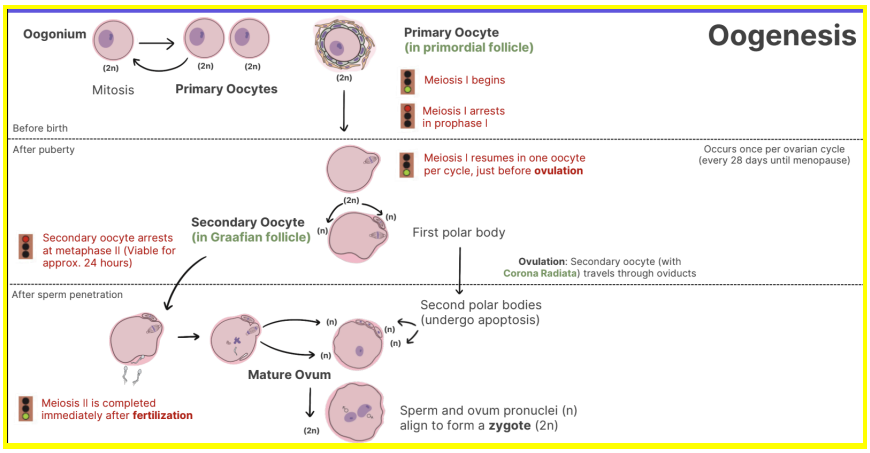


Associated with the **follicular phase** in the ovary

Driven by **estrogen** secretion

The **stratum functionalis** and the **stratum basalis** look similar: both tube small rings of glandular cells in cellular stroma.




Oogenesis
Oogonium
(2n)
Mitosis
(2n)
(2n)
Primary Oocytes
(2n)
Primary Oocyte
(in primordial follicle)
Meiosis I begins
Meiosis I arrests
in prophase I
Before birth
After puberty
Occurs once per ovarian cycle
(every 28 days until menopause)
Meiosis I resumes in one oocyte
per cycle, just before ovulation
(2n)
(n)
(n)
Secondary Oocyte
(in Graafian follicle)
First polar body
Secondary oocyte arrests
at metaphase II (Viable for
approx. 24 hours)
Ovulation: Secondary oocyte (with
Corona Radiata) travels through oviducts
After sperm penetration
Second polar bodies
(undergo apoptosis)
(n)
(n)
(n)
(n)
Mature Ovum
Meiosis II is completed
immediately after fertilization
(2n)
Sperm and ovum pronuclei (n)
align to form a zygote (2n)

1. Describe the mechanistic basis for the advantages of transcription-mediated amplification (TMA) over the polymerase chain reaction (PCR). BCM

Both are **NAATs** (nucleic acid amplification tests): they find a germ by making billions of copies of a piece of its genetic material until there is enough to detect. TMA is the preferred NAAT for chlamydia, gonorrhea, and trichomonas (vaginal swab or first-catch urine).

PCR (polymerase chain reaction)	**TMA (transcription-mediated amplification)**
Copies the germ's **DNA**. Each round has 3 steps at different temperatures: 1. **Heat (95 °C)** to split the DNA double helix into 2 single strands 2. **Cool** so **primers** (short pieces of DNA that mark where copying starts) stick to each strand 3. **DNA polymerase** (Taq) builds a new partner strand on each one Result: **2 copies** per round. Repeat 30–40 rounds by heating and cooling over and over.	Copies the germ's **rRNA** (ribosomal RNA). Everything happens at **one temperature (~41 °C)**: 1. A primer sticks to the rRNA. **Reverse transcriptase** builds a DNA copy of it 2. **RNase H** chews up the original RNA, leaving single-stranded DNA (no heat needed) 3. A 2nd primer makes it double-stranded DNA with a **T7 promoter** (a "start here" signal) 4. **T7 RNA polymerase** reads that DNA and makes **100–1,000 RNA copies**, and each copy goes back to step 1

Why TMA is better

	TMA	**PCR**	**Why TMA wins**
Starting target	rRNA: **thousands** of copies per cell (ribosomes)	DNA: **1 to a few** copies per cell	More to start with → **more sensitive**
Temperature	One temperature	Heat/cool cycles	No waiting to heat and cool → **faster**
Copies per round	**100–1,000**	**2**	~10^9 copies in under an hour
What it makes	**RNA**: breaks down quickly	DNA: stable, lingers in the lab	Leftover copies can't contaminate the next sample → **fewer false positives**

TLDR: TMA starts with more copies, makes more copies per cycle, and doesn't need heat cycling, so it is **more sensitive and faster**. Its RNA product doesn't stick around, so there are **fewer false positives**.

Describe the function of the estrogen and progesterone receptors and diagram their signaling pathways. BCM **Ishaan**

General mechanism for Androgen Receptors

Estrogen receptors (ER) and progesterone receptors (PR) are **Type I Nuclear Receptors** (these receptors are found in cytosol since steroid hormones can readily cross the plasma membrane)

Slow Genomic Signaling

- Inactive T1NHR bound to HSP90 → steroid hormone binding → nuclear localization → chaperone protein dissociation (HSP90) → homodimerization w/ another receptor-ligand complex → DNA binding via zinc fingers → transcriptional changes (via histone modification, transcription factor recruitment, etc.)

Rapid Genomic Signaling

- Binding to an AR receptor can induce a faster response that don't require transcriptional changes and protein synthesis by utilizing protein kinase cascades
    - E.g.s include MAPK/ERK, P13K/AKT, Src, and Ca++

Estrogen receptors:

- Alpha and beta isoforms with similar gene structure but are…
    - Different genes on different chromosomes
    - Different tissue/cell distribution
    - Different affinity for ligands
    - Different genes activated
- **ER alpha**: reproductive system
- **ER beta**: bone, brain, urinary tract
    -

Progesterone receptors:

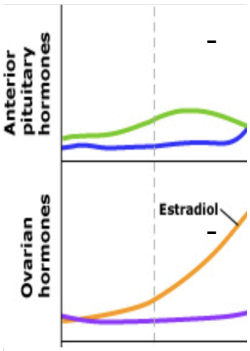
- Alpha and beta isoforms (**PR alpha** /**PR beta**)
- Differential transcriptional activities
- Cell type and promoter specific effects: often PR-beta > PR alpha
    - PR alpha can repress PR beta and ER signaling

Describe gonadotropin control of ovarian function and outline the phases of the ovarian cycle with the corresponding hormonal patterns. PHYSIO **Ishaan**

Follicular Phase (Days 1–14):

Two-Cells being stimulated

- LH acts on theca interna cells in ovarian follicles to induce the synthesis of androstenedione and progesterone from cholesterol
- FSH acts on granulosa cells to produce inhibin B and to convert androstenedione into estradiol via aromatase enzyme



Negative Feedback (Early Follicular)

- Increased circulating estrogen and inhibin B exert negative feedback on the anterior pituitary to lower FSH and LH levels → this starves immature follicles with lower FSH receptivity, driving them into follicular atresia while one dominant Graafian follicle emerges

Positive Feedback Switch (Late Follicular)

- As the Graafian follicle matures, high levels of estradiol reach a specific threshold
- This switches the pituitary response from negative to positive feedback, triggering a massive surge in LH secretion

Ovulation (Ovulatory Phase):

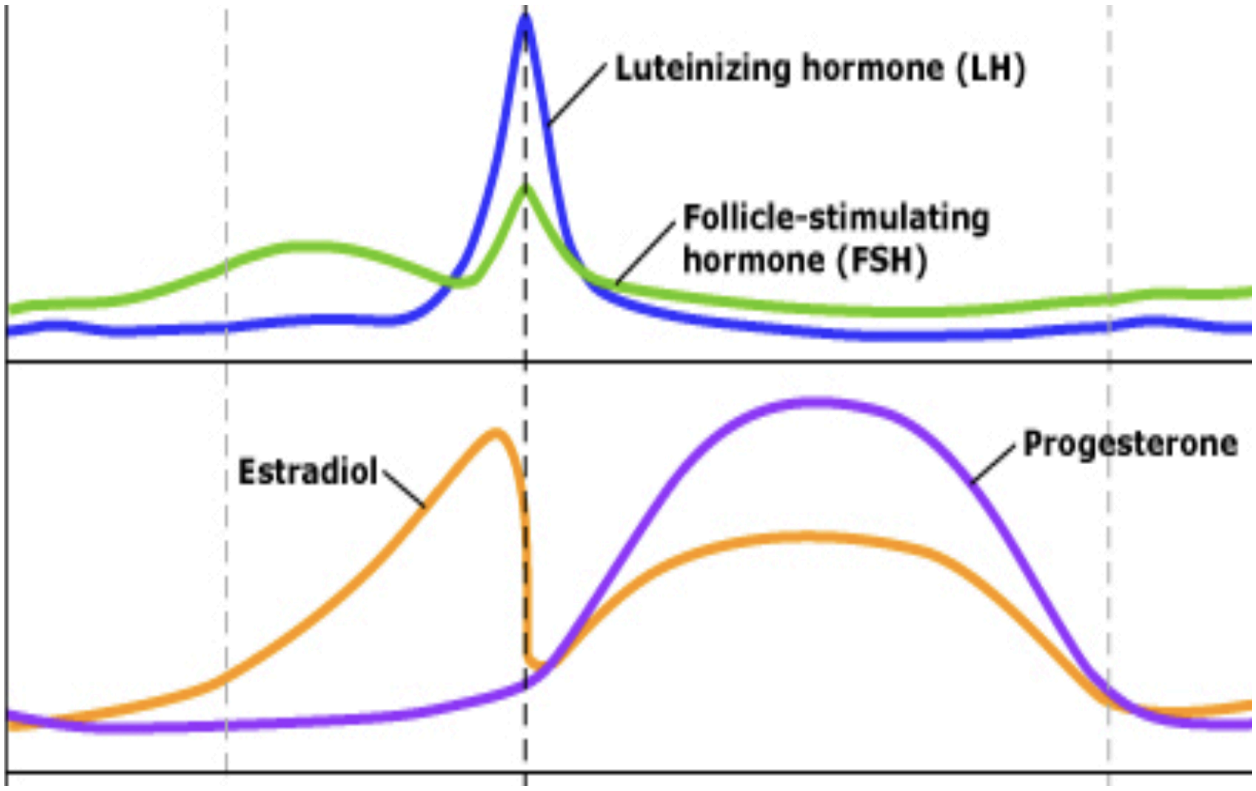
- The LH surge induces the dominant follicle to rupture, releasing the mature oocyte into the fallopian tube while also stimulating the oocyte to complete the final steps of meiosis I

Luteal Phase (Days 15–28):

- Following ovulation, LH transforms the remaining follicular cells into the corpus luteum (luteinization)
- Theca lutein and granulosa lutein cells secrete high levels of progesterone and estrogen

- Elevated progesterone and estrogen exert strong negative feedback on the hypothalamus and pituitary to suppress FSH and LH, preventing new follicles from maturing during this phase

- Menstruation: If fertilization does not occur, falling LH levels cause the corpus luteum to degenerate into the fibrous corpus albicans
- The resulting withdrawal of progesterone and estrogen leads to ischemia of endometrial spiral arteries and sloughing of the uterine wall, resulting in menses



Describe the principal physiologic actions of estrogen and progesterone on their target organs.
PHYSIO **Ishaan**

Estrogen		Progesterone
Promotes secondary sex characteristics at puberty: - pubertal growth spurt - pubic hair growth - breast development - female distribution of adipose iIncreases protein synthesis (including transport proteins like **SHBG**) Upregulates progesterone receptors	**Systemic & Anabolic Effects**	↑ body temperature ↑ breathing response to CO2 Depressant/ hypnotic effect on brain Inhibits GnRH release
- Development of female genitalia - Proliferation of endometrium - Stimulates thinner cervical mucus - Stimulates production of cervical mucus, making it less viscous - Increased myometrial excitability (contraction of uterus) - Stimulates growth of stromal cells in prostate - Essential for spermatogenesis	**Reproductive**	- Maintains pregnancy - Inhibits endometrial hyperplasia - converts the lining into a nutrient-rich, secretory tissue - thick cervical mucus - ↓ Estrogen receptor expression - smooth muscle relaxant - represses uterine contractions **so progesterone does not necessarily grow or degrade endometrium but prepares it for embryo*
Increases prolactin	**Breast**	Inhibits prolactin
- More HDL less LDL - Increased coagulation (less Antithrombin III, more II, VIII, IX, X, plasminogen)	**Blood**	- Less HDL more LDL -↑ basal insulin levels -↑ insulin response to glucose -↑ ketogenesis
Suppressed immune response	**Immune**	
Decreased bone resorption (inhibit osteoclasts)	**Bone**	

1. Describe the trigger for and physiologic process of normal menarche. PHYSIO **Shannon**

Yellow = Relevant to LO
Gray = Out of scope but probably useful for the exam

Menarche is the first menstrual period. It comes fairly late in puberty, after the hypothalamic–pituitary–ovarian (HPO) axis switches back on.

1. Background: why puberty is "on hold"

- GnRH neurons start in the **olfactory placode** and **migrate to the hypothalamus** in the fetus. Placental steroids suppress GnRH in utero, although LH and FSH reach near-adult levels at midgestation.
    - **Clinical link:** if this migration fails, the result is **Kallmann syndrome**
- After birth, the baby is cut off from placental and maternal steroids. This causes a brief **"mini-puberty"**: gonadotropins rise, with a short estradiol bump in XX infants.
- The axis then stays quiet through childhood for two reasons:
    - The hypothalamus and pituitary are **extremely sensitive to negative feedback** from even tiny amounts of sex steroids.
    - There is **CNS inhibition** of the GnRH pulse generator.
- Meanwhile, all oocytes sit as **primary oocytes arrested in prophase I** inside primordial follicles. They stay that way from before birth until puberty.
    - There are about 7 million germ cells at fetal month 5 and up to 2 million at birth. Fewer remain by puberty, and only about 400–500 will ever ovulate.

2. The trigger: the GnRH pulse generator matures

- **The key event:** CNS inhibition fades and feedback sensitivity drops. The hypothalamus then releases **pulsatile GnRH with greater amplitude and frequency**.
    - Pulses happen **at night (during sleep) first**, then around the clock.
    - Pituitary gonadotrophs become **more sensitive to GnRH**.
- **Upstream signals** *(standard Step 1)*:
    - **Kisspeptin** acts on GPR54 to stimulate GnRH neurons.
    - **Leptin** reflects fat stores and acts as a permissive "enough energy" signal.
    - This is why **low body fat** (athletes, anorexia) delays menarche, and **obesity** is linked to an earlier one. It also helps explain why the average age of menarche has fallen over time.
- **Step 1 point:** GnRH must be **pulsatile** to stimulate the pituitary. **Continuous** GnRH (leuprolide) suppresses it instead, which is the basis for treating central precocious puberty.

3. The hormone cascade

↑ pulsatile GnRH → ↑ **FSH and LH** from the anterior pituitary → ovaries respond:

- **FSH** drives follicle growth.
- **LH and FSH together** drive estradiol production (two-cell model):
    - LH acts on **theca cells**, which make androstenedione.
    - FSH acts on **granulosa cells**, which convert it to estradiol using **aromatase**.
- **Estradiol** drives the early changes: breast development and the growth spurt. The growth spurt needs **sex steroids plus GH, with GH acting through IGF-1**.
- **Adrenarche** (adrenal DHEA/DHEAS) causes pubic and axillary hair. It starts about **2 years before gonadal maturation** and runs independently of the HPO axis.

4. Physical sequence and timing

- **Order: thelarche (breast budding) → pubarche (pubic hair) → peak growth spurt → menarche**. Axillary hair and peak growth velocity follow within 1–2 years of the first signs.
- **Normal ages (US):**
  - Thelarche at about 10 years (usually 8–13).
  - Menarche at about **12–13 years**, roughly **2–3 years after thelarche**, around **Tanner stage 4**.
- Menarche comes **after peak height velocity**. Growth slows afterward because estrogen **closes the growth plates**.
- Estrogen at puberty also builds **peak bone mass**.

5. The first period itself

- Over time, rising estradiol **thickens the endometrium**: the stratum functionalis proliferates.
- **Mechanism of the first bleed** *(standard Step 1)*:
  - Early cycles are usually **anovulatory**. The positive-feedback LH surge is not yet mature, so there is no corpus luteum and no progesterone.
  - Menarche is therefore typically an **estrogen-withdrawal (or breakthrough) bleed**: estradiol fluctuates or falls, and the proliferated functionalis sheds.
- Later, the axis matures enough for estradiol to flip to **positive feedback**. That produces the LH surge, regular **ovulation**, a corpus luteum and progesterone, and true **ovulatory cycles**.
  - In an ovulatory cycle, menses happens when the corpus luteum regresses, progesterone drops, and the spiral arteries constrict. The resulting ischemia makes the functionalis slough off.
- **Pearl:** ovulation can happen **before** the first period, so a girl can get pregnant before menarche.

6. Definitions: when to worry

- **Precocious puberty:** secondary sexual characteristics **before age 8**.
- **Delayed puberty: no breast development by 13**.
- **Primary amenorrhea:** no menses **by 15** with normal secondary sexual characteristics, **or by 13 with no breast development**.
- **Normal cycle**: 24–38 days long, less than 8 days of bleeding, and less than 7–9 days of variation between cycle lengths.

7. The first years after menarche

- **Irregular, anovulatory cycles are normal for about 1–3 years.** They can cause heavy or irregular bleeding from **unopposed estrogen**, which over-thickens the endometrium.
- **But** heavy menses **since menarche**, plus a personal or family bleeding history, means **von Willebrand disease until proven otherwise**. Check a CBC, coagulation panel, and VWD testing.
- **Primary dysmenorrhea:**
  - It starts once cycles become **ovulatory**. Progesterone withdrawal leads to **prostaglandin** release, which causes uterine contractions and ischemia.
  - **Treatment:** NSAIDs first, then OCPs.
  - Severe or worsening pain should raise suspicion for endometriosis.

8. Long-term significance of the age at menarche

- **Early menarche, late menopause, and fewer pregnancies** all mean more lifetime estrogen exposure. That raises the risk of **breast and endometrial cancer** (and ovarian cancer, through more ovulations).
- **Delayed or absent puberty** (Turner, athlete triad, anorexia) leads to **low bone density** and fracture risk.

9. Primary amenorrhea: ***Extra but i found this helpful***

First step in any workup: rule out pregnancy with β-hCG. Next, check FSH, prolactin, and TSH (the same panel Cullen lists for ovulatory dysfunction).

Level	Cause	Key clues	FSH/LH
Hypothalamus	**Kallmann syndrome**: GnRH neurons fail to migrate	**Anosmia**, no puberty	↓
	Functional hypothalamic amenorrhea: low leptin from stress, excessive exercise, or anorexia	Female athlete triad (low energy availability, amenorrhea, **low bone density**)	↓
	Constitutional delay	Most common cause of delayed puberty; family history of late bloomers; delayed bone age	↓ (for age)
Pituitary	Prolactinoma or hypothyroidism	Prolactin suppresses GnRH; galactorrhea	↓
Ovary	**Turner syndrome (45,XO)**: streak ovaries ("no 2nd sexual maturation")	Short stature, webbed neck, coarctation, no breasts	↑↑ (hypergonadotropic hypogonadism)
Outflow tract	**Müllerian agenesis (MRKH)**	Normal breasts and pubic hair, normal ovaries, **absent uterus and upper vagina**, 46,XX	Normal
	Imperforate hymen or transverse vaginal septum	**Cyclic pelvic pain** with no bleeding, **bulging bluish hymen**, blood trapped in the vagina (hematocolpos)	Normal

Hormone receptor	**Androgen insensitivity (46,XY)**	Breasts present (testosterone is aromatized to estrogen), **scant or absent pubic and axillary hair**, no uterus, blind-ending vagina, testes in the abdomen or groin	Normal or ↑ LH

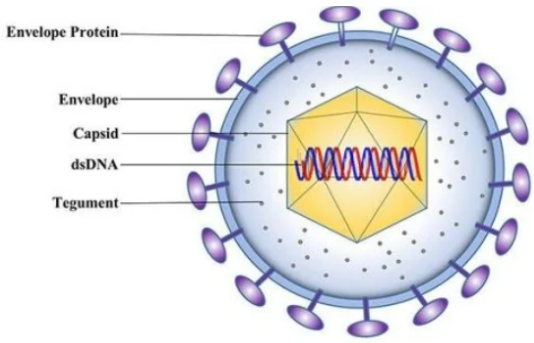
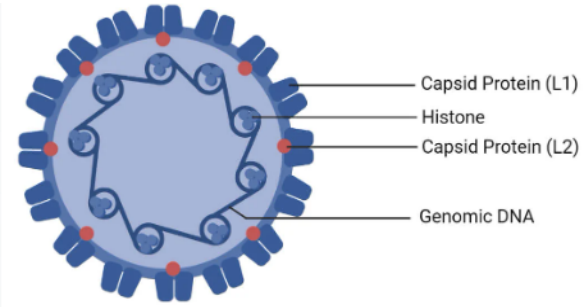
High-yield distinctions:

- **Breasts present** means estrogen is being made, so the problem is the outflow tract or a hormone receptor. **No breasts** means the problem is in the HPO axis.
- **High FSH** points to the ovary. **Low or normal FSH** points to a central (brain) cause.
- **MRKH vs. androgen insensitivity:** in MRKH, pubic hair is normal and the karyotype is XX. In androgen insensitivity, pubic hair is absent and the karyotype is XY.

10. Precocious puberty

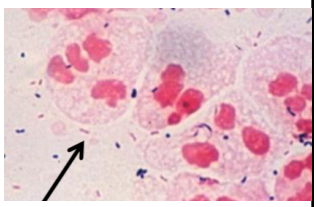
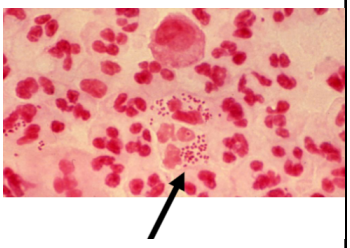
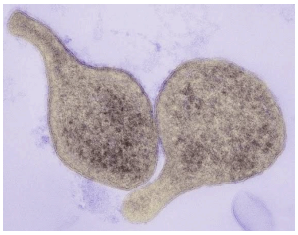
- **Central (GnRH-dependent):**
  - The pulse generator switches on early. In girls it is usually idiopathic.
  - Get a **brain MRI** to look for a CNS lesion, such as a hypothalamic hamartoma.
  - **Labs:** LH and FSH are high and rise further after a GnRH stimulation test.
  - **Treatment:** a **continuous GnRH agonist** (leuprolide).
- **Peripheral (GnRH-independent):**
  - The estrogen comes from outside the axis. Causes include a **granulosa cell tumor**, **McCune-Albright syndrome** (café-au-lait spots, polyostotic fibrous dysplasia, precocious puberty), and exogenous estrogen.
  - **Labs:** LH and FSH are **suppressed**.
- **CAH (21-hydroxylase deficiency):**
  - Excess adrenal androgens cause early pubarche or virilization .
  - Nonclassic CAH can look like PCOS, with irregular periods and hirsutism.
  - Check **17-OH progesterone**
- **Main consequence of untreated precocious puberty:** the growth plates fuse early. The child is tall at first but ends up **short as an adult**.
- **Benign variants:**
  - Isolated premature thelarche: breast budding only, with no growth acceleration.
  - Premature adrenarche: early DHEAS causing pubic hair or body odor, independent of the HPO axis.

8. Review the microbiology, pathogenesis, and clinical manifestations of herpes simplex virus and human papilloma virus. INIM JP

Category	HSV (Herpes Simplex Virus)	HPV (Human Papillomavirus)
Microbiology	Enveloped dsDNA virus (-1 and -2 are most pertinent to STI's) 	Non-enveloped DNA virus with ~40 mucosal strains (low-risk = 6, 11 vs. high-risk = 16, 18, 31, 33, 45, 52, 58) 
Pathogenesis	Infects skin and mucosal epithelial cells. Causes **primary infection** in seronegative individuals or establishes **latency in sensory nerve ganglia**, leading to **recurrent outbreaks** or **asymptomatic viral shedding**	Infects **basal mucosal epithelial cells**. High-risk oncogenic types **integrate viral DNA** into host genome, disrupting cell-cycle regulation and promoting epithelial cell dysplasia
Clinical Manifestations	Clusters of **painful vesicles** (small fluid-filled blisters) on **erythematous bases** (reddened underlying skin on penis, labia, perianal area, oral area, or buttocks) that **erupt, crust, and heal within 1 week**. Accompanied by **local lymphadenopathy** and **burning/stinging neuralgia**	- **Low-Risk**: cauliflower-shaped genital warts (**condylomata acuminata**) - **High-Risk**: frequently **asymptomatic initially**; causes **cervical/anogenital intraepithelial dysplasia** (precancerous mucosal cell changes, visible as **acetowhite lesions**)
Patient Complications	**Recurrent meningitis**, **encephalitis**, **keratitis** (corneal ulcers), **herpetic whitlow** (finger infection), **erythema multiforme** (target rash), **increased HIV acquisition**	**Squamous cell carcinoma (SCC)** of the cervix, vagina, vulva, penis, and oropharynx

9. Describe the microbiologic characteristics of the bacteria associated with urethritis and cervicitis (focus on Neisseria gonorrhea, Chlamydia trachomatis, Mycoplasma genitalium). INIM JP

I'm going to include syphilis (T. pallidum) here because we need to know it as a part of LO P10 anyways

Category	Chlamydia	Gonorrhea	Syphilis	M. Genitalium
Appearance				
Microbiology	Obligate intracellular, weakly gram-negative bacterium lacking a peptidoglycan cell wall	Gram-negative diplococci that reproduces via binary fission	Corkscrew-shaped spirochete (cannot be Gram-stained or cultured on routine media)	Flask-shaped bacteria (smallest known free-living organism!), lacks a peptidoglycan cell wall (resistant to beta-lactams, cannot gram stain)
Pathogenesis	Infectious non-replicating **elementary bodies (EBs)** enter columnar cells, convert to replicating **reticulate bodies (RBs)** within inclusions, reorganize to EBs, and exit via **cell lysis or extrusion**. All occurs within a **72-hour biphasic intracellular lifecycle**	**Pili** mediated mucosal attachment/invasion (**genes undergo recombination** to evade immunity). **Lipo-oligosaccharide (LOS)** incorporates **host sialic acid** to escape immune recognition	Enters via microabrasions, disseminates hematogenously early, and induces **obliterative endarteritis** with stage-dependent systemic tissue destruction	**Tip organelle** (**P140/P110 adhesins** encoded by *MG191/MG192*) and **wheel complex motor** drive **gliding motility** and tissue invasion. **Facultative intracellular** pathogen. Undergoes *Mgpar/Mgpa* gene recombination via **MG428 alternative sigma factor** for **antigenic and phase variation**
Clinical Manifestations	**Nongonococcal urethritis** (urethral inflammation without gonorrhea = **dysuria** +	**Urethritis** (urethral inflammation with **dysuria** and **profuse, thick purulent**	**Primary** = single/multiple **painless "punched-out"**	**Men** = acute & chronic/persistent **NGU** (urethral inflammation without gonorrhea =

	thin mucoid discharge), **cervicitis** (cervical lining inflammation + redness + **yellow mucopurulent discharge**). **LGV Serotypes L1–L3** (chlamydial lymphatic tissue infection) can have **painless genital ulcer** with **painful suppurative lymphadenopathy** (buboes), **frequently asymptomatic** especially in females	**discharge**), **cervicitis** (cervical lining inflammation + redness + **yellow mucopurulent discharge**), **pharyngitis**, **proctitis** (rectal lining inflammation), **conjunctivitis**, **frequently asymptomatic** especially in females	**chancre** (painless firm ulcer) with smooth raised borders) **Secondary** = disseminated **maculopapular rash (palms/soles)**, **condyloma lata** (broad moist warty lesions), **mucosal patches**, lymphadenopathy, and **alopecia** **Latent** = asymptomatic seropositivity (**Early <12 mos** vs **Late >12 mos**) **Tertiary** = **gummas** (destructive soft granulomatous tumors)	**dysuria** + **mucoid discharge**), **balanoposthitis** (glans and prepuce inflammation), **proctitis** (rectal lining inflammation) **Women** = mucopurulent **cervicitis** (cervical lining inflammation + redness + **yellow mucopurulent discharge**), **urethral syndrome** (painful urination with sterile pyuria)
Patient Complications	**Pelvic inflammatory disease** (ascending upper genital tract infection), **salpingitis** (tubal inflammation), **ectopic pregnancy**, **tubal sterility**, **Fitz-Hugh-Curtis syndrome** (liver capsule adhesions), **reactive arthritis triad** (arthritis, conjunctivitis, and urethritis/cervicitis), **trachoma** (blindness)	**Pelvic inflammatory disease** (ascending upper genital tract infection), **tubo-ovarian abscess**, **epididymitis** (epididymis tube inflammation), **tubal sterility**, **disseminated gonococcal infection (DGI)** (polyarthralgias, tenosynovitis, dermatitis, septic arthritis), **Fitz-Hugh-Curtis syndrome** (liver capsule adhesions)	**Cardiovascular complications** (aortitis, aortic aneurysm, bulging aorta), **neurosyphilis** (tabes dorsalis, general paresis, Argyll-Robertson pupil), **congenital defects** (snuffles, saddle nose, Hutchinson's teeth), **Jarisch-Herxheimer reaction** (antibiotic flu-like response)	**PID** ascending upper genital tract infection), **endometritis**, **salpingitis** (fallopian tube inflammation), **tubal factor infertility**, **spontaneous preterm birth**, **SARA** (post-STI joint inflammation), **multidrug resistance** (**23S rRNA** & **parC/gyrA** mutations)

10. Explain the pathogenesis, clinical manifestations and complications of infections affecting the vulva, vagina, cervix, and fallopian tubes, including pelvic inflammatory disease. INIM & PATH

Here is a handy summary table if you don't want to read the notes:

Feature	**PID**	**Trichomoniasis**	**BV**	**Candidiasis**
Primary process	Ascending infection	Protozoal infection	Dysbiosis	Fungal overgrowth
Discharge	Mucopurulent	Frothy, yellow-green	Thin, gray-white	Thick, clumpy white
Vaginal pH	Variable	>4.5	>4.5	≤4.5
Inflammation	Marked	Marked	Minimal	Marked
Hallmark	Cervical motion tenderness	Motile trophozoites + strawberry cervix	Clue cells + fishy odor	Budding yeast/pseudohyphae
Major complications	Infertility, ectopic pregnancy, TOA	HIV susceptibility, adverse pregnancy outcomes	STI susceptibility, obstetric complications	Recurrence, severe vulvitis

Pelvic Inflammatory Disease (PID): polymicrobial ascending infection of the upper female genital tract (endometrium, fallopian tubes, ovaries, and/or peritoneum). The main pathogen suspects are *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, with *Mycoplasma genitalium, BV-associated anaerobes, and endogenous vaginal flora* being less likely but still possible
Pathogenesis: Ascending infection through the endocervix → endometrium → fallopian tubes ± peritoneum. Epithelial invasion causes ciliary loss, cellular necrosis, purulent exudate, and fibrin deposition → inflammation, scarring, and adhesions
Clinical manifestations:

- Lower abdominal/pelvic pain, fever and malaise
- Cervical motion tenderness (Chandelier sign), uterine and/or adnexal tenderness
- Abnormal vaginal/menstrual bleeding
- Mucopurulent cervical discharge

Complications:

- Tubal scarring/adhesions can lead to impaired gamete transport, tubal infertility, and ectopic pregnancy risk
- Tubo-ovarian abscess (encapsulated purulent collection involving the tube and ovary)
- Fitz-Hugh–Curtis syndrome: perihepatitis with violin-string adhesions between liver capsule and abdominal wall → RUQ pain that may have right shoulder referral
- Peritonitis

Trichomoniasis (*Trichomonas vaginalis*): sexually transmitted, flagellated anaerobic protozoan; four anterior flagella + recurrent flagellum associated with an undulating membrane → characteristic jerky motility on saline wet mount. Exists only as a trophozoite (no cyst stage) with mainly sexual transmission.
Pathogenesis: trophozoite adhesion to vaginal/urethral squamous epithelium → micro-ulceration, epithelial desquamation, tissue injury, and PMN recruitment
Clinical manifestations:

- Frothy, yellow-green discharge with vulvovaginal pruritus/burning
- Elevated vaginal pH (5–6)
- Strawberry cervix (punctate cervical hemorrhages, occurs approximately 2–5%)
- Males are frequently asymptomatic and may have mild urethritis (discharge/dysuria)

Complications:

- Mucosal inflammation leads to increased HIV acquisition/transmission risk
- Associated with preterm birth, PROM, and low birth weight.
- Elevated risk of PID, particularly in women with HIV
- Possible associations with infertility and cervical abnormalities

Bacterial Vaginosis (BV): polymicrobial vaginal dysbiosis characterized by depletion of protective lactic acid–producing **Lactobacillus spp.** and overgrowth of anaerobic/facultative organisms, especially *Gardnerella vaginalis, Fannyhessea vaginae (formerly Atopobium vaginae), Mobiluncus, Prevotella, Bacteroides, Mycoplasma hominis, and Ureaplasma spp.*
Pathogenesis: altered vaginal microbiota (associated with douching, sexual activity, etc.) → *Gardnerella*-associated epithelial biofilm → anaerobic expansion and production of volatile amines (putrescine, cadaverine) → fishy odor, elevated pH, and epithelial exfoliation. Minimal PMN response distinguishes BV from inflammatory vaginitis
Clinical manifestations (need Amsel criteria ≥3 of 4):

1. Thin gray-white discharge coating vaginal walls
2. Vaginal pH >4.5
3. Positive whiff test (fishy amine odor after adding 10% KOH)
4. Clue cells (squamous epithelial cells coated with bacteria, ≥20% on wet mount)

Complications:

- Associated with late miscarriage, PROM, preterm birth, and postpartum/postabortion endometritis
- Increased susceptibility to HIV, gonorrhea, chlamydia, HSV-2, and HPV acquisition/persistence
- Associated with PID and postoperative pelvic infections (ex: vaginal cuff cellulitis after hysterectomy)

Vulvovaginal Candidiasis: opportunistic overgrowth of *Candida spp.,* predominantly *C. albicans* (budding yeast, pseudohyphae, true hyphae). Non-albicans species include *C. glabrata, C. krusei, C. tropicalis, and C. parapsilosis.* Asymptomatic vaginal colonization is common (~10–20% or more, depending on population).
Pathogenesis: altered vaginal flora/host defenses → Candida proliferation, epithelial invasion (particularly hyphal C. albicans), and PMN-mediated inflammation

- Risk factors: broad-spectrum antibiotics, diabetes, immunosuppression, pregnancy/elevated estrogen

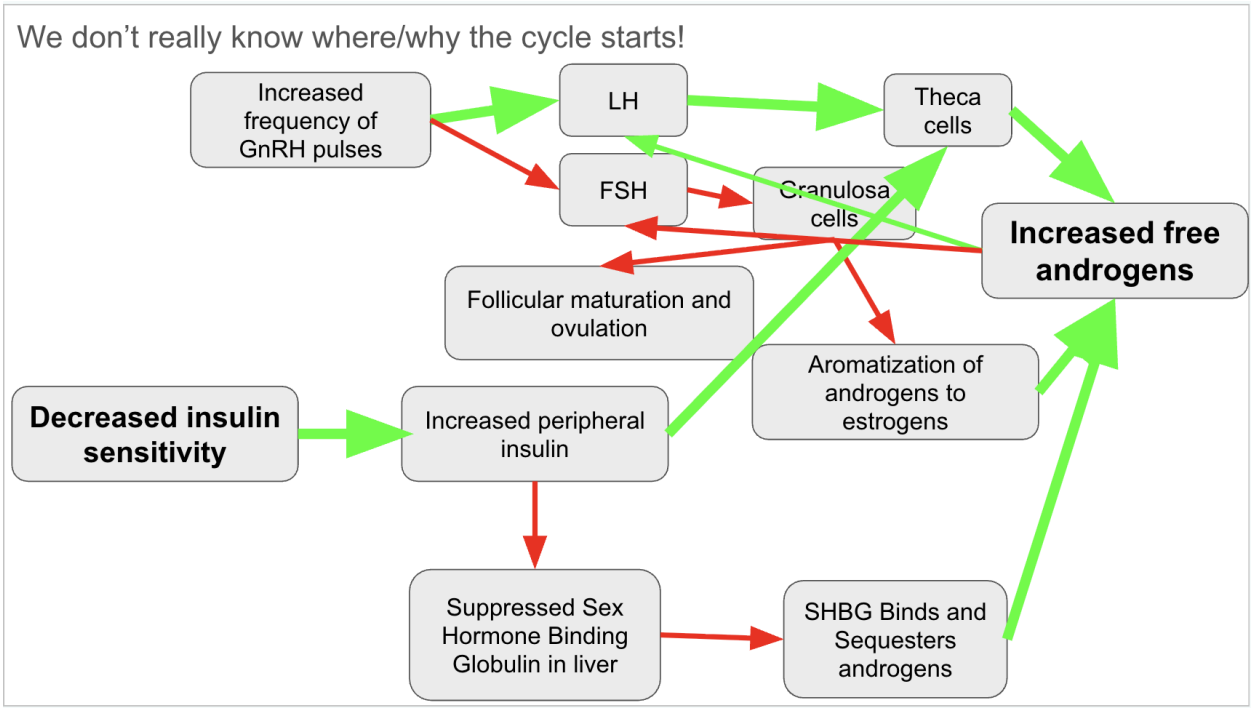
Clinical manifestations:

- Thick, white, clumpy cottage cheese-like discharge
- Intense vulvovaginal pruritus, burning, dyspareunia, erythema, edema, excoriations
- Normal vaginal pH (≤4.5)
  Budding yeast ± pseudohyphae/hyphae with inflammatory cells on KOH microscopy. C. glabrata may lack visible hyphae

Complications:

- Recurrent VVC: ≥3 symptomatic episodes within 12 months, may require prolonged induction and maintenance therapy
- Severe inflammation → vulvar fissures, excoriations and potentially secondary bacterial infection
- Non-albicans Candida is often less responsive to fluconazole
    - Species-directed management may require extended non-fluconazole topical azoles or selected intravaginal boric acid regimens

1. Explain the pathophysiologic basis of polyendocrine metabolic ovarian syndrome (PMOS), including the contributions of hyperandrogenism, disrupted gonadotropin secretion, and insulin resistance to its clinical presentation. PATH **Sophie**



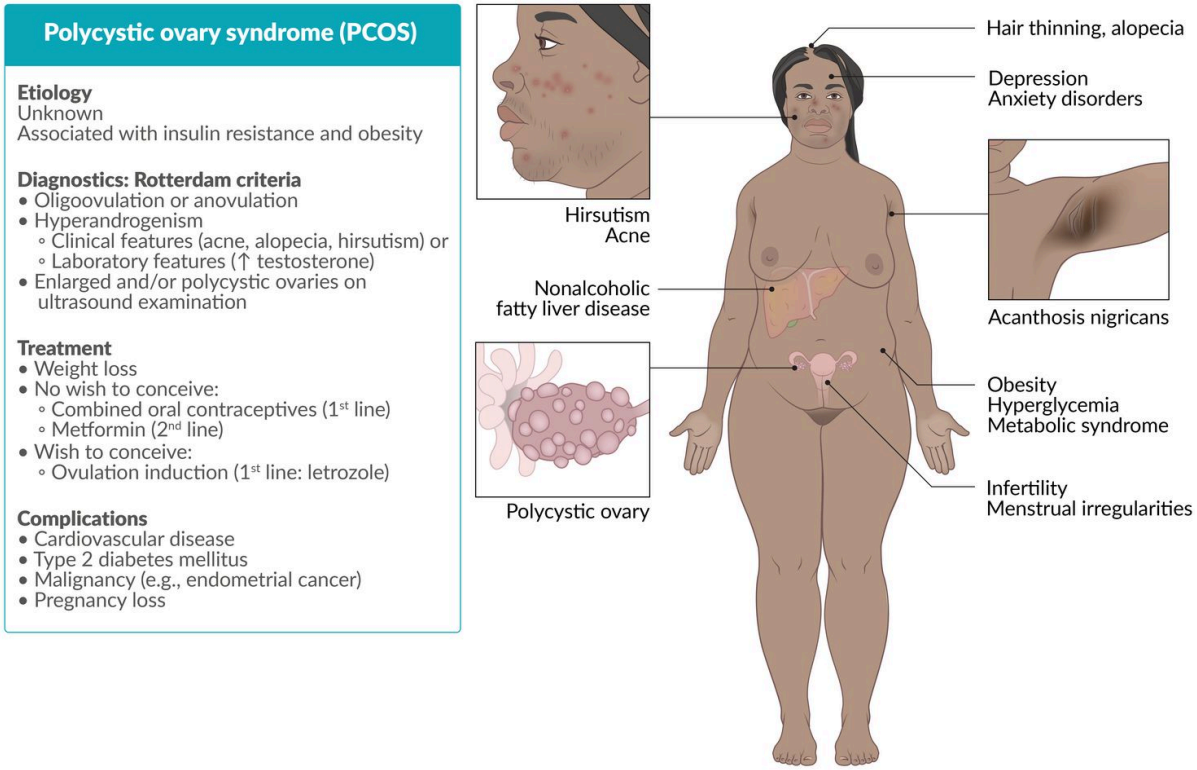
Unclear the etiology of the syndrome or why these things become dysregulated; however, it is clear how these things feed into each other

- Increased frequency of **GnRH** pulses in the hypothalamus increases **LH** secretion from the pituitary (**FSH** production relatively low compared to LH)
    - **LH** release stimulates **theca** cells in the ovaries which…
        - increases **androgen** production
    - Relatively low **FSH** levels provide little stimulation to **granulosa** cells which…
        - Does not aromatize (much) androgen to estrogens
            - Increases **androgen** androgen production
- Simultaneously, there is decreased insulin sensitivity
    - Thus more insulin is produced and less insulin is taken up into cells-> increasing **peripheral insulin**
        - Increased peripheral insulin…
            - Directly increased free **androgen** production
            - Suppresses **sex hormone binding globulin**
                - Less things binding estradiol and testosterone leading to more biologically active **free androgens**

<u>Clinical presentations:</u>

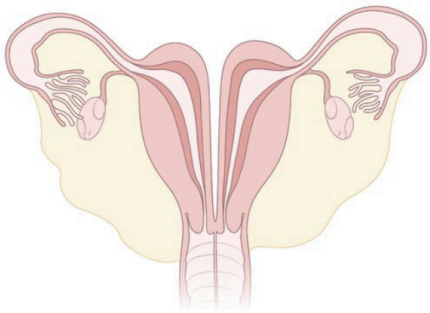
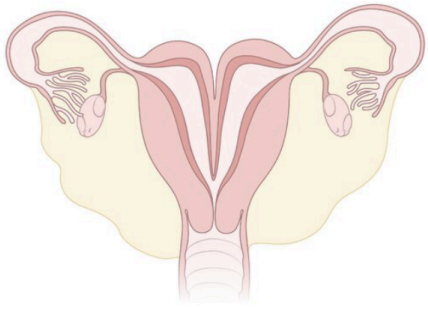
- Low FSH

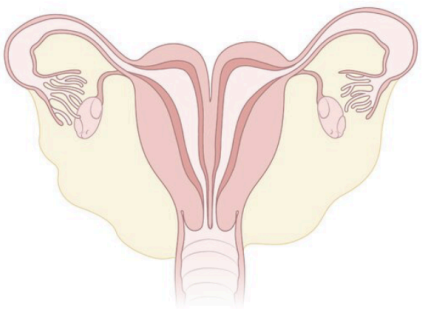
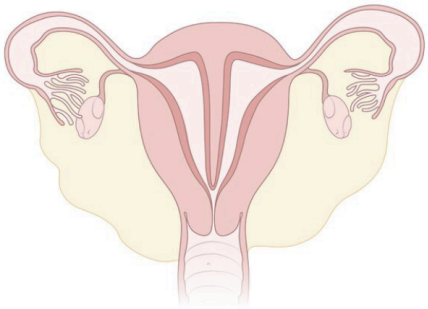
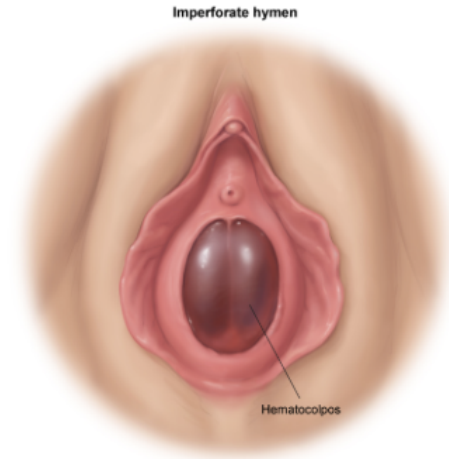
    - Does not stimulate follicular maturation and ovulation
        - Amenorrhea
        - Anovulation, oligovulation
            - Lack of shedding of endometrium can lead to endometrial hyperplasia if chronic
- Increased free androgens
    - Cystic acne
    - Hirsutism
    - Androgenic alopecia
- Metabolic syndrome
    - Decreased insulin sensitivity
        - Acanthosis nigricans
        - Other diabetes symptoms
    - High cholesterol
    - HTN
    - Dyslipidemia
    - Obesity
- OTher
    - Depression
    - Anxiety
    - Infertility

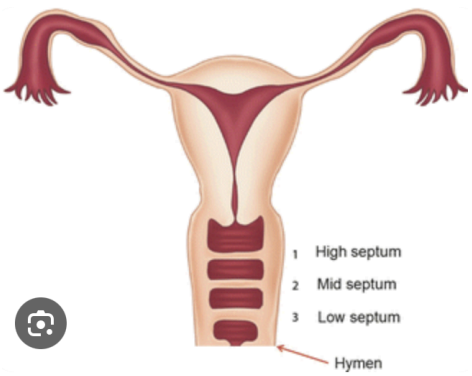
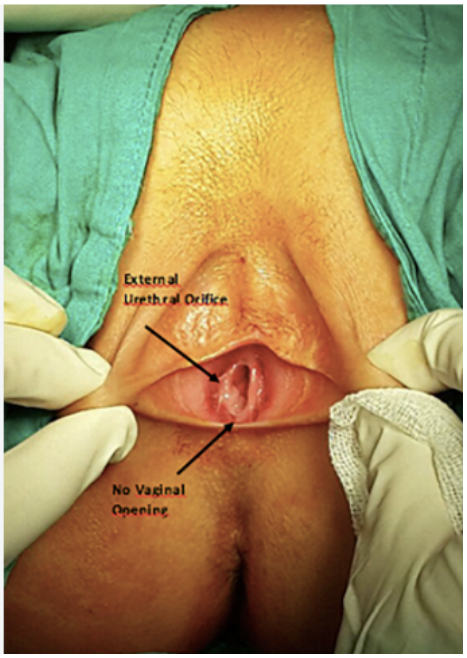


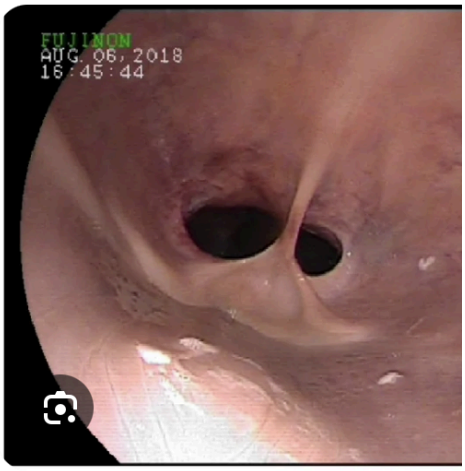
1. Compare and contrast common malformations of the Müllerian system and lower reproductive tract, and identify those that may present as primary amenorrhea.

The major distinction is where development failed: the Müllerian ducts form the fallopian tubes, uterus, cervix, and upper vagina, whereas the lower vagina and hymen involve the sinovaginal bulbs, vaginal plate, and urogenital sinus.

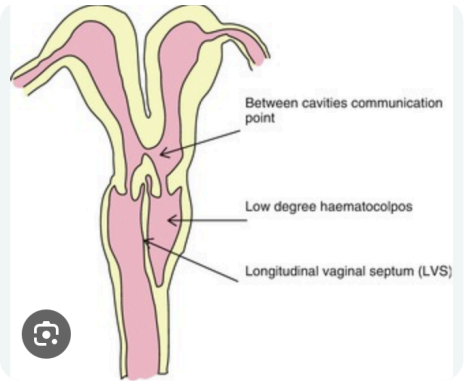
Mullerian System Malformations			
	Mechanism	Anatomy & presentation	Primary amenorrhea
Mullerian Agenesis	Bilateral failure of mullerian duct dvt	Absent or hypoplastic uterus, cervix, and upper vagina; ovaries are functional; normal secondary sexual characteristics and puberty	Yes. 15% of all amenorrhea
Unicornate uterus	One mullerian ducts doesn't develop	Single uterine horn, sometimes with a noncommunicating rudimentary horn. Img below	Usually no unless vagina is also absent
Uterus didelphys	Complete lack of Müllerian duct fusion → double uterus, double cervix, double vagina	Two uteri, two cervices, and often a vaginal septum  Didelphic uterus (double uterus and double vagina)	Usually no
Bircornuate uterus	Incomplete fusion of the Müllerian ducts to various degrees ● Uterus bicornis unicollis: double uterus, single cervix, and single vagina ● Uterus bicornis bicollis: double uterus and double cervix with/without a vaginal septum	Two uterine horns; one or two cervices, depending on subtype  Uterus bicornis unicollis	Usually no

		Arcuate uterus  Uterus bicornis bicollis	
Septate uterus	• The Müllerian ducts fuse, but the septa between the two ducts persists either partially (subseptate uterus) or completely (septate uterus).	One external uterus divided by an intrauterine septum  Septate uterus	Usually no
Lower Tract Malformations			
Imperforate hymen	Hymenal membrane fails to perforate - Central cells of the Müllerian eminence in the urogenital sinus do not disintegrate →imperforate hymen → cryptomenorrhea at puberty (outflow tract obstruction leads to backup of menstrual blood) → hematocolpos (accumulation of blood in the vagina) and/or hematometra (accumulation of blood in the uterus)	Bulging bluish membrane, hematocolpos 	Yes, but usually brought in by cyclic pain
Transverse vaginal septum	Vertical fusion failure. Most often in the upper third of the vagina - Failure of recanalization of	- Episodic pelvic pain, - hematocolpos due to Cryptomenorrhea, - Dyspareunia: pain w sex	Yes(obstructed menses)

	the Müllerian duct → transverse septum in the upper-third(45%), lower third (15–20%), and/or middle third (35–40%) of the vagina		
Lower vaginal agenesis	abnormal development of the sinovaginal bulbs and vaginal plate - Cervix fails to develop or its canal is obstructed - Uterus may be present, but menstrual blood cannot exit normally.	Absent lower vagina with a normal uterus, cervix, upper vagina, and ovaries; cyclic pain may occur 	Yes
Vaginal septum	Persistent longitudinal or transverse partition due to Abnormal fusion or canalization of the vagina - longitudinal septum divides the vaginal canal and may coexist with uterine didelphys - But a complete obstructing transverse septum may cause primary amenorrhea and cyclic pain.	Dyspareunia, obstructed flow, or occasionally normal menstruation depending on patency	Sometimes

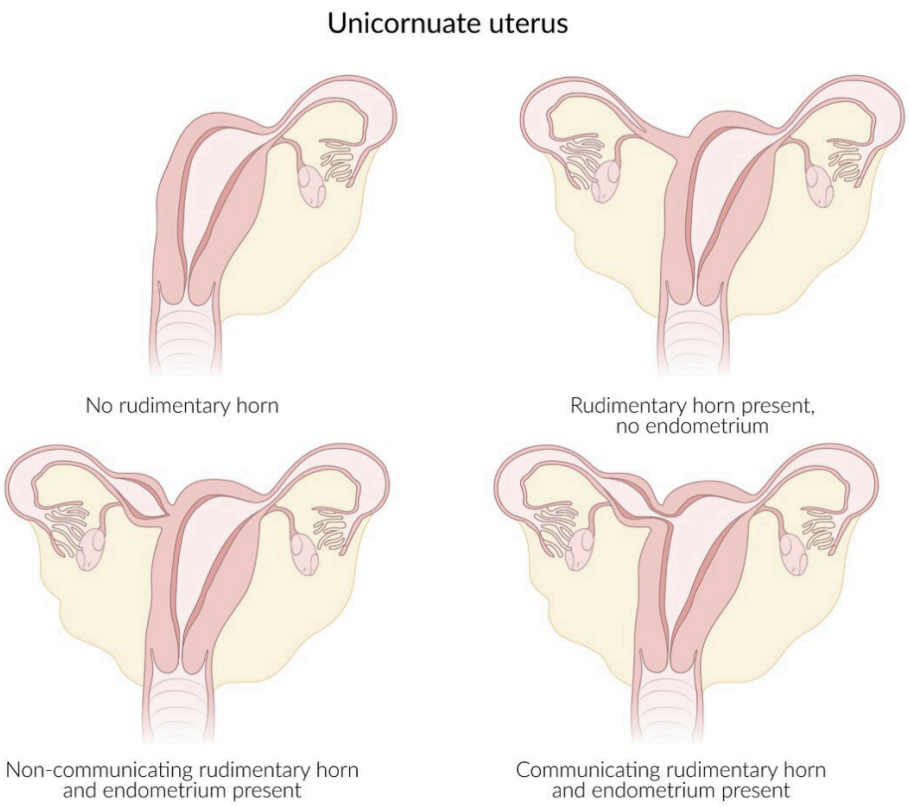


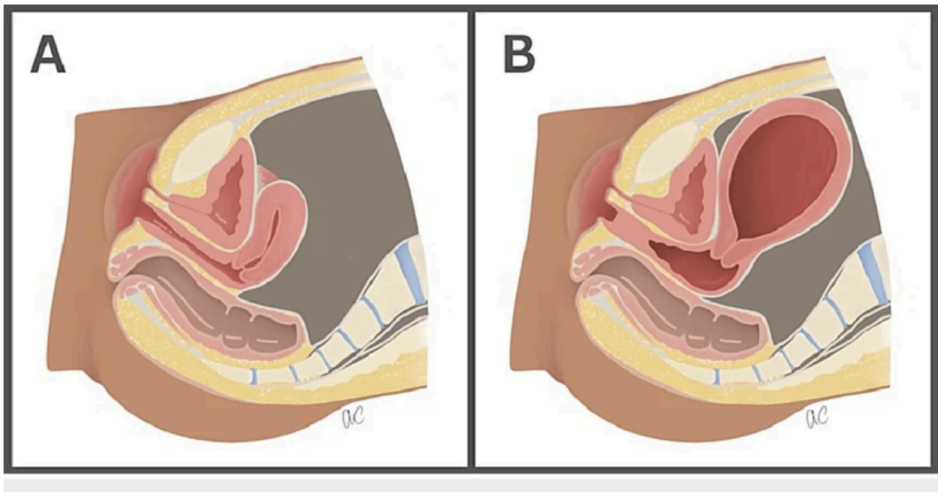
Endoscopic appearance of a longitudinal vaginal septum as...



Vaginal Septae and Imperforate Hymen | Obgyn Key

Visit >





A) Normal vaginal anatomy hemipelvis. B) Vaginal agenesis of the lower vagina with retained blood products and normal external vaginal anatomy hemipelvis. Illustration by Ashley Calise.

Describe and apply the components of the Tanner stages. CS

- Breast development (B1–B5) in girls.
- Genital development (G1–G5) in boys.
- Pubic hair development (PH1–PH5) in both sexes
    - PH6 is sometimes used to describe further hair growth in boys.

Stage	**Genital development (G)**	**Breast development (B)**	**Pubic hair (PH)**
1 — Prepubertal	Testes, scrotum, and penis have a prepubertal appearance and size.	No breast development.	Usually no pubic hair.
2 — Early puberty	Testes enlarge to approximately 4–5 mL; scrotum enlarges and its skin may redden. Penile growth has not yet begun.	Breast bud forms beneath the areola; areola enlarges slightly.	Sparse, lightly pigmented hair at the base of the penis.
3 — Mid-puberty	Testes and scrotum continue enlarging; penile growth begins.	Further breast enlargement; breast tissue extends beyond the areolar diameter.	Darker, coarser, curlier hair spreads over the pubic region.
4 — Late puberty	Testes and scrotum enlarge further; scrotal skin darkens; penis increases in length and diameter, and the glans develops.	Areola and nipple form a secondary mound above the breast contour.	Adult-type hair that does not extend onto the inner thighs.
5 — Mature	Testes, scrotum, and penis attain adult appearance and proportions.	Adult breast contour; areola generally returns to the breast contour, with the nipple projecting.	Adult-pattern hair extending onto the inner thighs.

Breast development

Stage 1

Elevation of papilla only

Stage 2

Breast bud: elevation of breast and papilla as a small mound and enlargement of areolar diameter

Stage 3

Additional enlargement of breast and areola with no separation of their contours

Stage 4

Areola and papilla project from surface of breast to form secondary mound

Stage 5

Mature stage with projection of papilla only with recession of the areola to the general contour of the breast

Female pubic hair development

Stage 1

The vellus over the pubes is the same as that over the anterior abdominal wall

Stage 2

Sparse slightly pigmented, downy hair along the labia that is straight or only slightly curled

Stage 3

Hair spreads sparsely over the pubic region and is darker, coaser and curlier

Stage 4

Hair is adult type, but the area covered is smaller than in most adults, and there is no spread to the medial surface of the thighs

Stage 5

Hair is adult in quantity and type, distributed as an inverse triangle and spreads to the medial surface of the thighs but not up the midline anterior abdominal wall

Male pubic hair and genital development

Stage 1

Penis, testes, and scrotum are the same size and proportion as in early childhood

The vellus hair over the pubic region is the same as that on the abdominal wall

Stage 2

Testes and scrotum enlarge and scrotal skin shows a change in texture and reddening

Sparse growth of straight or slightly curled pigmented hair appearing at the base of the penis

Stage 3

Penile growth in length more than width; further growth of the testes and scrotum

Hair is coarser, curlier, and darker; spread sparsely over the junction of the pubes

Stage 4

Further penile growth and development of the glans; further enlargement of testes and scrotum

Adult type hair, but area covered less than in most adults; no spread to the medial surface of the thighs

Stage 5

Genitalia are adult in size and shape

Adult in quantity and type of hair, distributed as an inverse triangle; spread is to the medial surface of the thighs

2. Compare and contrast the history and physical exam elements in primary versus secondary amenorrhea. CS - Ayla P

The history and physical exam for primary and secondary amenorrhea share many features. The H&P for patients with primary amenorrhea should focus more on growth history and pubertal development, while the H&P for patients with secondary amenorrhea should focus more on sexual history and past menstrual cycles.

Patient history

Primary amenorrhea	**Both**	**Secondary amenorrhea**
- Growth history	- HPI - Family history - Medical history - Medications - Assess for pituitary/hypothalamus symptoms: headache, vision changes, galactorrhea - Assess for thyroid symptoms: heat/cold tolerance, weight changes, energy levels - Diet/exercise habits and life stress	- Thorough sexual and reproductive history, including characteristics of past menstrual cycles

Physical exam and workup

Primary amenorrhea	**Both**	**Secondary amenorrhea**
- Assess growth parameters and pubertal development (Tanner stage)	- Vital signs - Pelvic exam assessing normal anatomy - Abdominal exam - Skin exam: assess for acne, hirsutism - Thyroid exam	- Assess body habitus

1. Categorize the causes of primary amenorrhea by the following etiologies: hypothalamic/pituitary disease, ovarian disease, uterine/outflow tract disorders, receptor/enzyme deficiencies. CM - Ayla P

HPO axis disfunction

- Pituitary tumor (especially prolactinoma): Can suppress LH/FSH production via structural compression, hyperprolactinemia, or both
- TBI: causing hypothalamic and/or pituitary dysfunction
- Cushing syndrome: elevated cortisol suppresses GnRH release
- Functional hypothalamic amenorrhea: excess stress or energy insufficiency suppresses HPO axis

Ovarian disease

- Pure gonadal dysgenesis: caused by failure of primitive gonads (ovaries/testes) to mature, resulting in “streak” gonads
    - Mullerian/Wolffian structures generally develop normally
- Premature ovarian insufficiency: loss of ovarian function before age 40
    - Most cases (80-90%) are idiopathic
    - Also associated with Turner Syndrome, Fragile X Syndrome, autoimmune diseases

Uterine/outflow tract disorders

- Mullerian agenesis: failure of Mullerian structures (uterus, uterine tubes, cervix, upper vagina) to develop
    - Ovaries usually intact and functional
- Transverse vaginal septum: transverse wall preventing passage through vaginal canal
- Imperforate hymen: thin mucosal membrane that covers the entire vaginal opening, causing obstruction of outflow

Receptor/enzyme deficiencies

- Kallman syndrome: genetic disorder (hereditary or sporadic) causing impaired GnRH release from hypothalamus
    - Causes hypogonadotrophic hypogonadism; generally diagnosed in adolescence when puberty fails to occur
    - Often accompanied by hyposmia/anosmia
- Androgen insensitivity syndrome: 46,XY karyotype with mutation in androgen receptor gene
    - External genitalia appear female, but Mullerian structures and ovaries absent
    - Internal undescended testes present

2. Compare and contrast primary and secondary amenorrhea and the general diagnostic approach to evaluation. CM - Ayla P

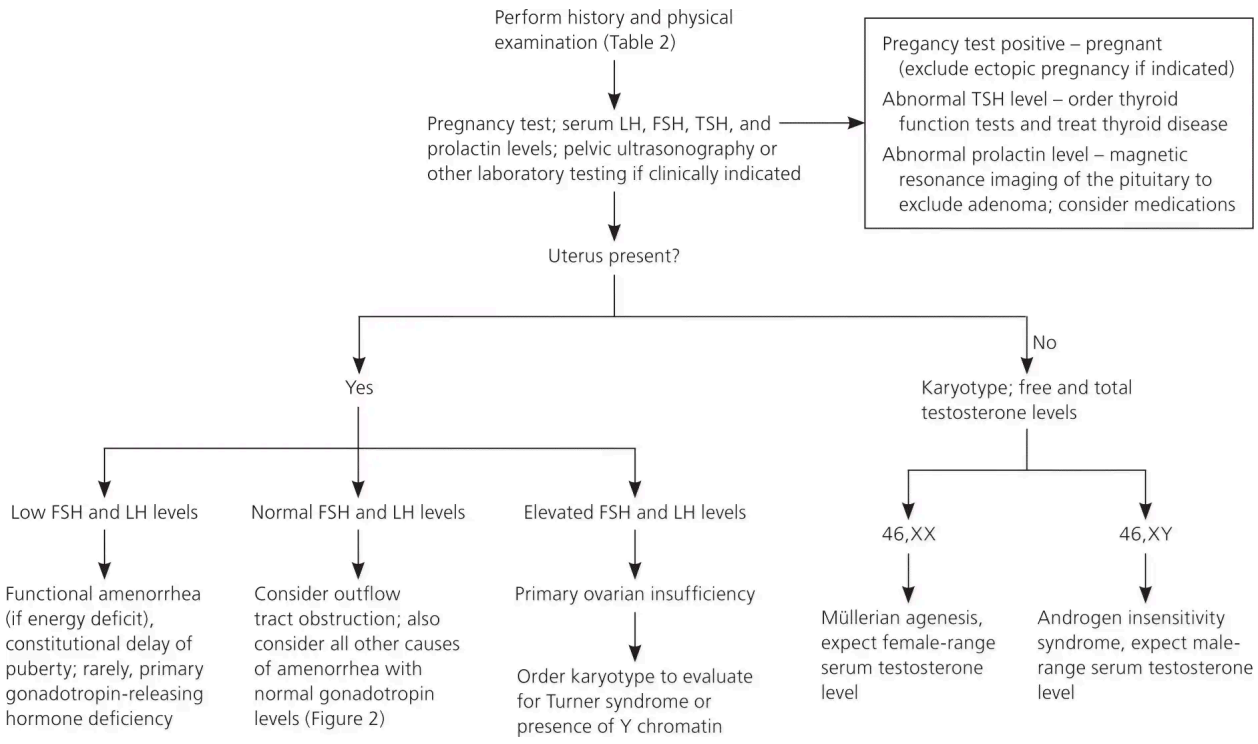
	Primary amenorrhea	**Secondary amenorrhea**
Definition	absence of a menstrual period past age 15 if other puberty changes are present, OR absent past age 13 if there is a complete absence of secondary sexual development	Absence of menstrual periods for 3+ consecutive months in a person who has previously had regular cycles, or for 6+ months in someone with irregular cycles
Prevalence	~0.1-0.3%	3-4%
Common causes	Ovarian insufficiency (Turner syndrome, gonadal dysgenesis, idiopathic) Functional hypothalamic amenorrhea Outflow tract anomalies	**Pregnancy** PCOS/PMOS Functional hypothalamic amenorrhea Thyroid disorders Medications

Diagnostic approach

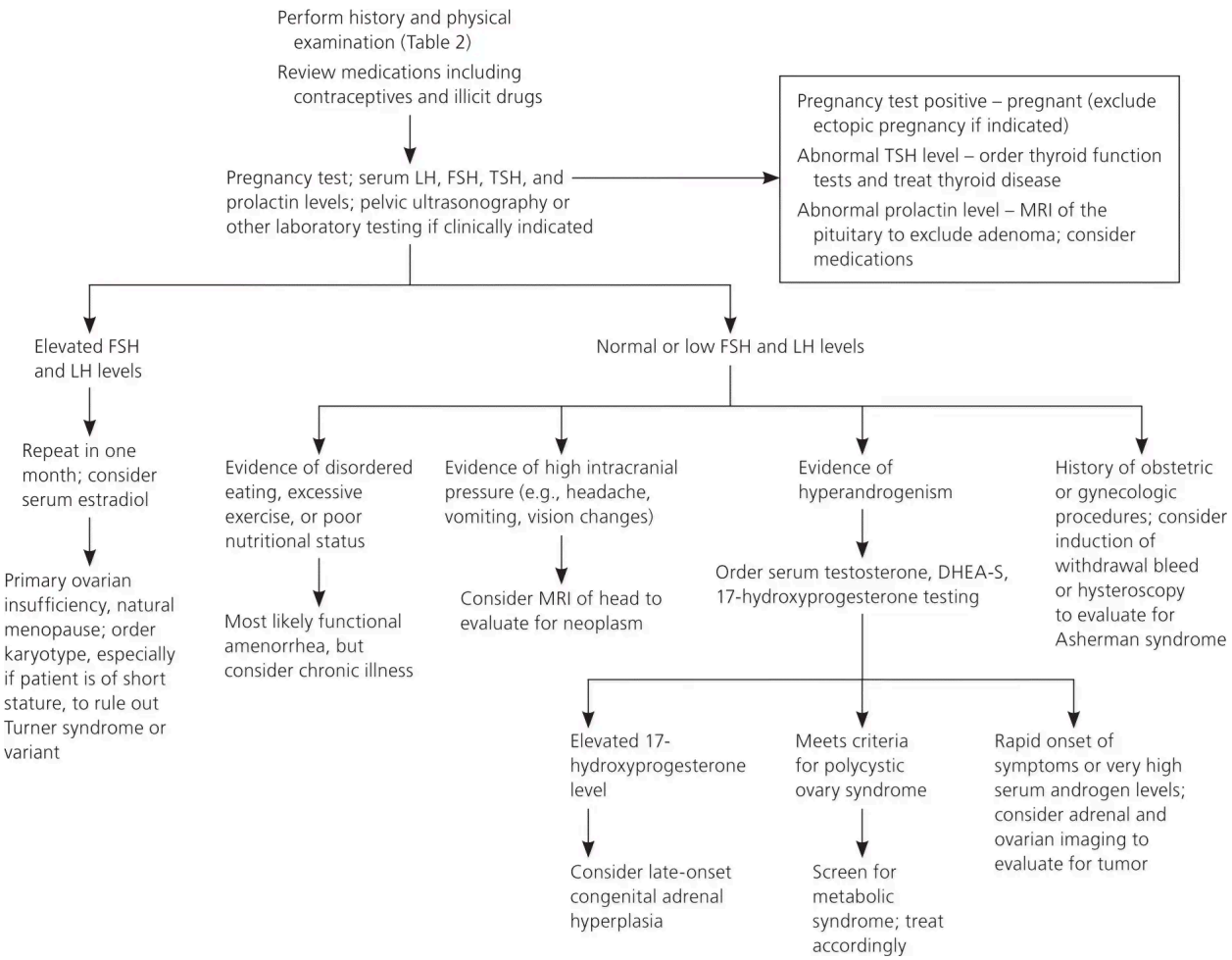
1. Identify primary vs secondary amenorrhea
2. Thorough H&P (See CS LO #1)
3. Labs: pregnancy test, CBC, CMP, baseline hormonal profile (TSH, LH, FSH, estradiol, testosterone, prolactin)
4. Based on findings, other labs/imaging may be done:

- Pelvic ultrasound: to assess for masses or abnormal anatomy
- Brain MRI: to assess for pituitary tumors
- A1C: if PMOS/PCOS is suspected
- Karyotype: in cases of primary amenorrhea, if a chromosomal/genetic cause is suspected

Flowchart (primary amenorrhea)



Flowchart (secondary amenorrhea)


Perform history and physical examination (Table 2)
Review medications including contraceptives and illicit drugs
Pregnancy test; serum LH, FSH, TSH, and prolactin levels; pelvic ultrasonography or other laboratory testing if clinically indicated
Pregnancy test positive – pregnant (exclude ectopic pregnancy if indicated)
Abnormal TSH level – order thyroid function tests and treat thyroid disease
Abnormal prolactin level – MRI of the pituitary to exclude adenoma; consider medications
Elevated FSH and LH levels
Repeat in one month; consider serum estradiol
Primary ovarian insufficiency, natural menopause; order karyotype, especially if patient is of short stature, to rule out Turner syndrome or variant
Normal or low FSH and LH levels
Evidence of disordered eating, excessive exercise, or poor nutritional status
Most likely functional amenorrhea, but consider chronic illness
Evidence of high intracranial pressure (e.g., headache, vomiting, vision changes)
Consider MRI of head to evaluate for neoplasm
Evidence of hyperandrogenism
Order serum testosterone, DHEA-S, 17-hydroxyprogesterone testing
History of obstetric or gynecologic procedures; consider induction of withdrawal bleed or hysteroscopy to evaluate for Asherman syndrome
Elevated 17-hydroxyprogesterone level
Consider late-onset congenital adrenal hyperplasia
Meets criteria for polycystic ovary syndrome
Screen for metabolic syndrome; treat accordingly
Rapid onset of symptoms or very high serum androgen levels; consider adrenal and ovarian imaging to evaluate for tumor

Describe the screening recommendations and available diagnostic tests for chlamydia, gonorrhea, trichomonas, HSV and syphilis. INIM

Key Points

- NAAT is preferred for CT, GC, trich and HSV lesions.
- Syphilis uses two-step serology: treponemal plus nontreponemal.
- No routine HSV serologic screening.
- Trich screening is mainly for women with HIV.

Screening by Population

- Women <25 y: CT/GC annually (USPSTF B). Syphilis if at risk.
- Women ≥25 y: CT/GC if at increased risk (new or multiple partners, partner with an STI, transactional sex).
- Men who have sex with women: CT/GC evidence is insufficient (USPSTF I). Not screened for trich.
- **MSM:**
    - CT/GC at least annually at all exposed sites (urethral, rectal, pharyngeal)
    - Syphilis at least annually
    - Every 3–6 mo if higher risk
- **HIV:**
    - CT/GC and syphilis at entry to care, then at least annually
    - Trich annually if receptive vaginal sex
- **Pregnancy:**
    - Syphilis for all at the first prenatal visit, plus 28 wk and delivery if high risk
    - CT/GC if <25 y or at risk. Retest CT in the 3rd trimester.
    - Trich at the first visit if HIV-positive
- **PrEP:**
    - GC every 6 mo and CT every 12 mo (MSW and women). MSM and transgender women: every 6 mo, including extragenital sites.
    - Syphilis every 6 mo (every 3–4 mo if high risk)
- **Repeat testing:**
    - Multiple or anonymous partners → every 3–6 mo
    - Any positive result → rescreen at 3 mo (reinfection is common)
    - Pharyngeal GC → test of cure at 7–14 d

Diagnostic Tests

- **Chlamydia/Gonorrhea**
    - NAAT: one specimen tests for both
    - Women: vaginal swab (self-collected is OK). Men: first-void urine. Add rectal and pharyngeal swabs by exposure.
    - Culture for GC susceptibility, e.g., when a test-of-cure NAAT is positive

- **Trichomonas**
  - NAAT preferred (95–100% sensitivity)
  - Wet mount sensitivity is only 44–68%, falling to about 20% after 1 hr. A negative wet mount → NAAT.
  - Point of care: OSOM rapid antigen (82–90%) or GeneXpert (<1 hr)
- **HSV**
  - Lesion PCR preferred (97–100%): unroof the vesicle and swab the base
  - PCR gives the type: HSV-1 recurs less often than HSV-2
  - Culture is less sensitive (30–70%)
  - Type-specific IgG (glycoprotein G) if no lesions or for partner counseling
    - Antibodies appear 2 wk–3 mo after infection
    - HSV-2 index <3.0 → confirm
  - No IgM testing
- **Syphilis**
  - Traditional: RPR/VDRL → confirm with treponemal test (TP-PA, FTA-ABS)
  - Reverse: treponemal EIA/CIA → RPR
    - Better for early and latent disease, but more false positives
    - Discordant results → second treponemal test
  - Treponemal tests stay positive for life
  - RPR/VDRL titers track treatment. A 4-fold rise means reinfection.
  - Darkfield and NAAT are not widely available
  - Treat suspected syphilis at the first visit
- **Also:** any genital ulcer or new STI → test for **HIV** and screen for other STIs

1. Describe and identify the morphology of endometrial hyperplasia and endometrial carcinoma; explain the pathogenesis of these conditions, including why a patient with PMOS is at increased risk for their development.

	Endometrial Hyperplasia	Endometrial Carcinoma
Pathogenesis	- Due to prolonged unopposed estrogen stimulation w/out adequate progesterone opposition - Estrogen drives endometrial proliferation - Progesterone converts proliferative enzyme into secretory endothelium and limits further growth	- Type I endometrial cancer is the most common type; endometrioid adenocarcinoma (grades 1 and 2) , which typically develops through prolonged estrogen exposure and often arises from atypical hyperplasia. - Type II endometrial hyperplasia: endometrioid adenocarcinomas (grade 3) and tumors of nonendometrioid histology (serous, clear cell, mucinous, squamous, transitional, and undifferentiated cells) Lymphogenic and hematogenous metastases can be seen in advanced stages of endometrial cancer.
Morphology	**Endometrial hyperplasia without atypia** (benign endometrial hyperplasia) - Simple endometrial proliferation - Few cysts - Enlarged, dilated mucous membrane glands (swiss cheese pattern) btw **abundant** stromal tissue - No cytological atypia (eg enlarged, irregular nuclei & loss of normal polarity) - Low risk of progressing to carcinoma **Endometrial hyperplasia with atypia** (endometrial intraepithelial neoplasia; EIN) - Pronounced proliferation of glandular tissue - Glands positioned back to back, no separating stroma - Clonal, atypical cells with extensive mitosis and loss of cell polarity - **Very high risk of endometrial carcinoma**	Type 1: - Pronounced gland proliferation, presents as atypical glandular tubes - Glands positioned back-to-back, almost no stroma - Lined with pseudostratified epithelial cells, the nuclei of which are enlarged in an atypical vesicular form. - These glandular cells frequently demonstrate mitosis. - Tumor cell nests may also be observed and infiltrate the myometrium in high-grade tumors. Type 2: - Papillary or glandular architecture - Severe nuclear pleomorphism w frequent mitoses

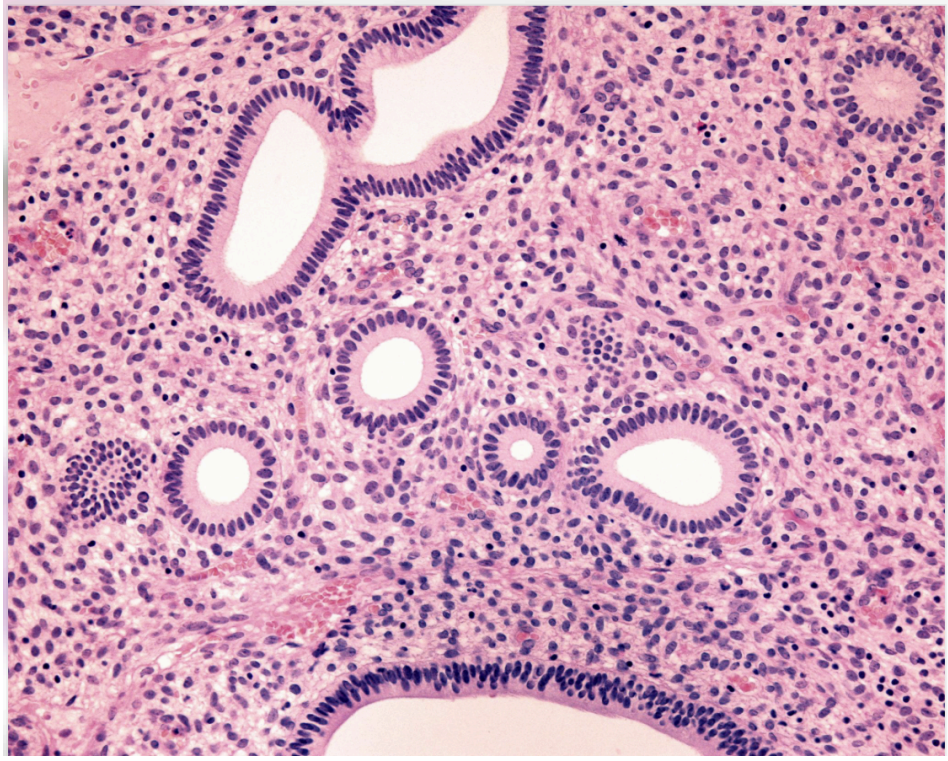
	- Cysts can still be present in atypical hyperplasia/EIN, but they don't define it and don't distinguish it from benign hyperplasia.	

Why does PMOS increase risk?

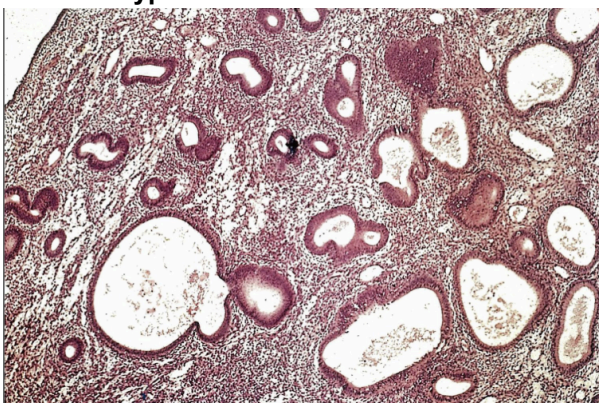
- Chronic anovulation: no corpus luteum means no cyclic progesterone. Persistent follicular arrest keeps estradiol production going, causing prolonged unopposed proliferation without regular secretory transformation or shedding.
- Hyperandrogenism: elevated testosterone drives follicular arrest. The endometrium expresses androgen receptors and 5α-reductase, and receptors are overexpressed in some women with PMOS. Genetically higher testosterone is linked to endometrial cancer. LH hypersecretion may also contribute.
- Obesity (more than 50% of patients): adds peripheral conversion of androgens to estrogens and increases androgen production.

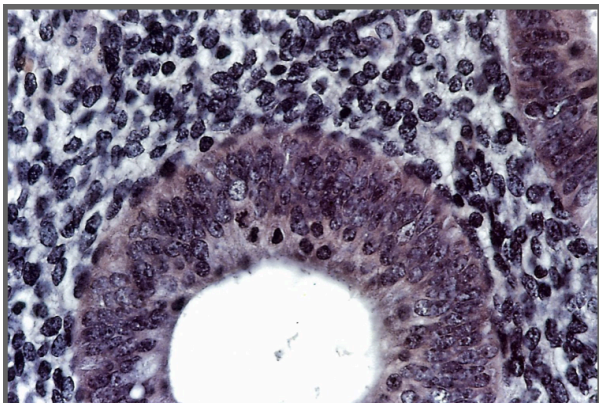
Overall, the biggest contributor is prolonged exposure to estrogen w/out the progesterone buffer.
PCOS → chronic anovulation → decreased progesterone → unopposed estrogen → endometrial hyperplasia/EIN → increased risk of endometrioid carcinoma.

Normal Endometrium:



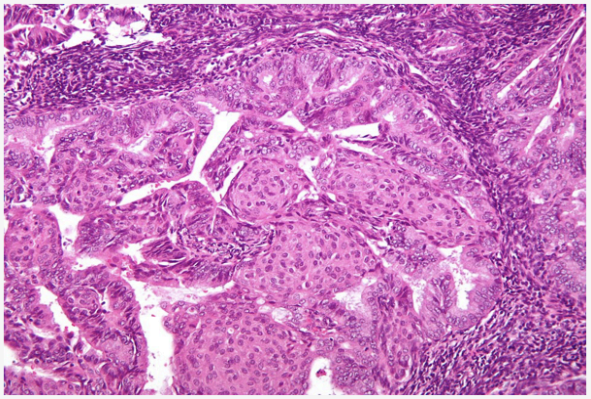
Without Atypia:

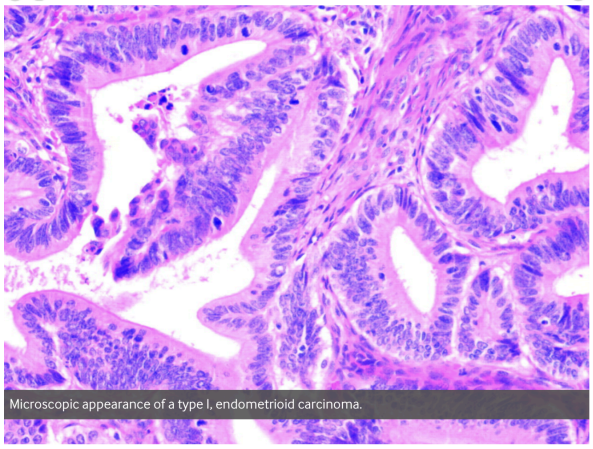




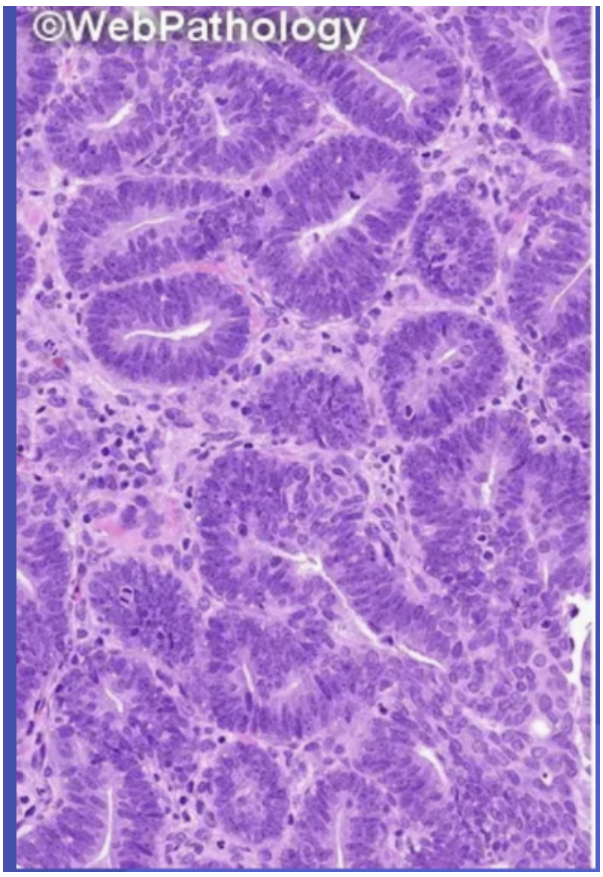
magnified image shows columnar epithelium with no mitoses or suspected neoplastic cells (normal finding)

With Atypia:





Both T1 - I'll try and find a typ2 img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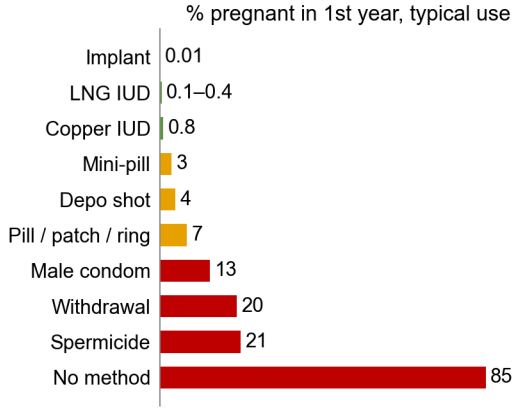
1. Compare their efficacy, complications, and contraindications for contraceptive options. CM and PHARM **Ravi**



- **Typical use** = real life (missed pills, late shots)
- **Perfect use** = used correctly every time
- **Pearl Index** = pregnancies per 100 woman-years
- **LARCs** (implant, IUDs) are >99% and don't depend on the user
- **Combined (E + P):** stop ovulation. Progestin blocks the LH surge (main effect); estrogen ↓ FSH and stabilizes the endometrium
- **Progestin-only:** thick cervical mucus ± stop ovulation
- **Copper IUD, barriers:** act on sperm; condoms are the only STI protection

Methods

	Method	Fail %	How it works	Side effects	Contraindications
	Implant Nexplanon (etonogestrel), arm; 3 yr (works 5)	0.01	• ↓ Ovulation • Thick mucus • Thin endometrium	• **Irregular, unpredictable bleeding** • ↓ Dysmenorrhea	• OK if estrogen is contraindicated
	LNG IUD Mirena, Liletta 8 yr; Kyleena 5 yr; Skyla 3 yr	0.1–0.4	• Local only: thick mucus, ↓ sperm motility, thin endometrium • **Ovulation still happens**	• Spotting and cramps for 3–6 mo, then **lighter periods** • Treats heavy, painful periods • **Protects against endometrial hyperplasia**	• IUD contraindications
	Copper IUD Paragard, no hormones; 10 yr (works 12)	0.8	• Sterile inflammation toxic to sperm • ↓ Tubal motility	• **Heavier, crampier periods** • Keeps a normal monthly cycle • Best emergency contraception	• IUD contraindications • **Wilson disease, copper allergy**
	Depo shot DMPA, IM every 3 mo	4	• ↓ Ovulation • Thick mucus	• Irregular bleeding, then **amenorrhea** (40–50% at 1 yr) • **Reversible ↓ bone density** • **Weight gain** (only method that causes it)	• OK if estrogen is contraindicated
	Combined pill Daily; 21 active + 7 placebo	7	• E + P: ↓ ovulation • Thick mucus, ↓ tubal motility	• **VTE**, stroke, MI, HTN • 1st-pass liver metabolism • Less acne and hirsutism (↑ SHBG)	• Estrogen contraindications
	Patch Weekly ×3, 1 wk off	7	• Same as pill	• No 1st pass → 60% more estrogen → **highest VTE risk** • Less effective if >90 kg	• Estrogen contraindications • Weight >90 kg
	Vaginal ring 3 wk in, 1 wk out; Annovera lasts 1 yr	7	• Same as pill	• **Lowest estrogen dose** • No nausea, least breakthrough bleeding	• Estrogen contraindications
	Mini-pill Norethindrone, norgestrel (Opill, OTC), drospirenone	3	• Thick mucus • Drospirenone also ↓ ovulation	• **Must take within the same 3-hr window** • Irregular bleeding • Drospirenone: ↑ **K+**	• OK if estrogen is contraindicated • Drospirenone: hyperkalemia risk

Contraindications

No estrogen if:	No IUD if:
• Smoker age ≥35 • Hypertension • Migraine with aura • History of DVT/PE or clotting disorder (factor V Leiden) • Stroke or ischemic heart disease	• Pregnancy • Active PID, cervicitis, or vaginitis • Unexplained vaginal bleeding • Cervical or endometrial cancer • Distorted uterus (fibroids, anomalies) • Copper IUD only: Wilson disease, copper allergy

Why VTE: estrogen makes the liver produce more clotting factors (V, VIII, X, fibrinogen) and less anticoagulant protein. Risk is highest in the first 3 months.

Hormone side effects

	Estrogen	Progestin
Too much	Nausea, breast tenderness, headache, bloating	↑ Appetite and weight, acne, hirsutism, low mood
Too little	**Early-cycle spotting (days 1–9)**, no withdrawal bleed	Heavy periods, cramps, **late-cycle spotting (days 10–21)**

Emergency contraception

	Method	Use within	How it works	Notes
	Copper IUD	5 days	Toxic to sperm, ↓ tubal motility	**Most effective**; 52 mg LNG IUD works too
	Ulipristal (ella)	5 days	Progesterone receptor modulator, delays ovulation	Rx only; weaker if BMI >35
	Levonorgestrel (Plan B)	3 days	Blocks LH surge, delays ovulation	OTC; weaker if BMI >30

Oral EC prevents fertilization, not implantation. It does not end an existing pregnancy and has no contraindications besides allergy.

Describe the treatment strategies for the common STIs (empiric vs targeted therapy and treatment of contacts). CM

*** I think days of treatment are relevant, don't think doses really matter, but basically just know HIGH DOSE = shorter duration, LOWER DOSE = longer duration ***

Key Points (2021 CDC)

- Gonorrhea: ceftriaxone 500 mg IM alone. Azithromycin co-treatment is no longer recommended.
- Chlamydia: doxycycline 100 mg BID × 7 d at any site.
- PID: ceftriaxone + doxycycline + metronidazole (anaerobes)
- Syphilis, GC and CT are reportable in all states.
- Partner notification for every STI diagnosis, except when there is intimate partner violence

Empiric Therapy (treat at the visit, before NAAT results)

- Cervicitis/urethritis: doxycycline, plus ceftriaxone if GC risk is high
- Proctitis: ceftriaxone + doxycycline × 7 d. Doxycycline × 21 d if LGV is suspected.
- PID: ceftriaxone + doxycycline + metronidazole × 14 d
- Suspected early syphilis: benzathine penicillin G (BPG) at the first visit
- Known contacts: treat presumptively

Targeted Therapy (after NAAT confirms the organism)

- Ceftriaxone alone if CT is excluded. Add doxycycline if it isn't.
- Don't treat asymptomatic PCR-detected colonizers.

Targeted Regimens (nonpregnant adults)

- **Gonorrhea - ceftriaxone**
  - Ceftriaxone 500 mg IM once (1 g if ≥150 kg)
  - Cephalosporin allergy → gentamicin 240 mg IM + azithromycin 2 g PO
  - No ceftriaxone → cefixime 800 mg PO (unreliable for pharyngeal infection)
  - Pharyngeal → test of cure at 7–14 d
- **Chlamydia - doxycycline**
  - Doxycycline 100 mg BID × 7 d
  - Alternatives: azithromycin 1 g once (weaker for rectal infection) or levofloxacin × 7 d
  - Pregnancy: azithromycin, then test of cure at 3–4 wk
- **Trichomonas - metronidazole**
  - Women: metronidazole 500 mg BID × 7 d
  - Men: metronidazole 2 g once
  - Alternative: tinidazole 2 g once
- **HSV** (not curative) - **valacyclovir**

    - First episode: valacyclovir 1 g BID, acyclovir 400 mg TID, or famciclovir × 7–10 d
    - Recurrent, episodic: valacyclovir 500 mg BID × 3 d
    - Suppression: valacyclovir 500 mg daily (1 g daily if ≥10 recurrences/yr)
- **Syphilis - Penicillin G**
    - Primary, secondary or early latent: penicillin G 2.4 MU IM once
        - Alternative: doxycycline × 14 d
    - Late latent or unknown duration: penicillin G 2.4 MU IM weekly × 3
        - Alternative: doxycycline × 28 d
        - Pregnancy: a gap >9 d between doses means restarting the series
    - Neuro, ocular or otic: aqueous penicillin G IV × 10–14 d
    - Pregnancy: penicillin only (desensitize if allergic)
    - Penicillin allergy: <1% is true IgE-mediated, so evaluate with skin testing and an amoxicillin challenge
    - Jarisch-Herxheimer reaction in up to 50–75% of patients treated for early syphilis

Follow-Up

- Syphilis: RPR/VDRL every 3–6 mo
    - Expect a 4-fold decline by 12 mo for early syphilis
    - By 24 mo for latent syphilis or HIV
- GC/CT: retest at 3 mo (reinfection up to 12%)

Treatment of Contacts

- CT/GC: treat all partners from the past 60 days, or the most recent partner if there was no sex in that window
- Expedited partner therapy (EPT): treat partners without examining them
    - Doxycycline for CT and cefixime 800 mg for GC
    - Packaged medication is preferred over prescriptions, with educational materials
    - For GC, try to link partners to in-person ceftriaxone first
    - EPT lowers reinfection compared with patient referral
- MSM: EPT is a shared decision, because HIV coinfection could be missed. Assess HIV-negative partners for PrEP.
- Syphilis and trichomonas: treat contacts presumptively. Syphilis partners are usually handled by public health partner services.

1. Describe the preferred treatment regimens, their mechanism of action, and adverse effects for Neisseria gonorrhea and Chlamydia trachomatis as recommended by the CDC. PHARM **Shannon**

Gonorrhea (*N. gonorrhoeae*)
Preferred: Ceftriaxone, one IM dose.

- 500 mg if under 150 kg; 1 g if 150 kg or more.
- **If chlamydia hasn't been ruled out, add a chlamydia drug:**
  - Doxycycline 100 mg twice daily for 7 days if not pregnant.
  - ★ **Azithromycin instead if pregnant.**
- The CDC no longer adds azithromycin routinely, because gonorrhea resistance to it is rising.
- ★ Resistance has been described and is getting worse (Weber).

Alternatives:

- **Cephalosporin allergy:** gentamicin 240 mg IM once plus azithromycin 2 g by mouth once.
- **Ceftriaxone unavailable:** cefixime 800 mg by mouth once. It's unreliable for pharyngeal gonorrhea.
- **Pharyngeal gonorrhea:** ceftriaxone is the only recommended option. Do a test of cure 7–14 days after treatment.

Chlamydia (*C. trachomatis*)
Preferred: doxycycline 100 mg by mouth twice daily for 7 days.

- It works at genital, rectal ★ **and throat** sites.
- ★ Rectal chlamydia is common in women and can't be predicted from the sexual history a patient reports. Azithromycin works less well there.
- No resistance has been described.
- ★ **If gonorrhea results aren't back yet, also give ceftriaxone 500 mg IM once** (Weber).

Alternatives:

- **Azithromycin** 1 g by mouth once, for when a patient may not finish a 7-day course (e.g., some teenagers); giving the dose in clinic helps. ★ Because it works less well for rectal infection, testing after treatment may be needed.
- ★ **Delayed-release doxycycline** 200 mg once daily for 7 days. It works as well and causes fewer GI side effects, but costs more.
- **Levofloxacin** 500 mg daily for 7 days. It works but is more expensive.
- ★ **Erythromycin is no longer recommended.** Its GI side effects make patients stop taking it.
- **Pregnancy:** azithromycin is preferred, with amoxicillin 500 mg three times daily for 7 days as the alternative.

★ **Step 1 link:** Chlamydia lacks a classic peptidoglycan cell wall. Why ceftriaxone alone doesn't cover it and you need a 30S or 50S protein-synthesis inhibitor. Amoxicillin in pregnancy is the clinical exception.

Mechanisms and adverse effects

Drug	Mechanism	Key adverse effects

Ceftriaxone (3rd-gen cephalosporin, β-lactam)	Binds penicillin-binding proteins and blocks peptidoglycan cross-linking, so the cell wall fails and the bacterium **lyses** → bactericidal	Injection-site pain and redness, diarrhea and nausea, rash/allergy, *C. difficile*, eosinophilia, ★ **leukopenia**, raised liver enzymes. ★ Most penicillin-allergic patients tolerate it because its side chains differ.
Cefixime (oral 3rd-gen cephalosporin)	Same as ceftriaxone	Mostly GI. ★ Overall side-effect rate is similar to ceftriaxone.
Doxycycline (tetracycline)	Binds **30S** and blocks aminoacyl-tRNA attachment → bacteriostatic	Nausea and vomiting, **pill esophagitis or ulcers** (worst if taken right before bed), **photosensitivity**, **fetal tooth discoloration** (avoid in pregnancy). ★ Also: rash, rarely SJS/TEN, liver toxicity, raised BUN, hemolytic anemia, low platelets.
Azithromycin (macrolide)	Binds **50S** and blocks translocation. Its long intracellular half-life allows a single dose.	GI upset, **QT prolongation**, liver toxicity, drug interactions. ★ Rising resistance in *N. gonorrhoeae* and *M. genitalium*.
Gentamicin (aminoglycoside)	Binds **30S** and causes mRNA misreading → bactericidal	**Kidney damage and ototoxicity**, both uncommon after one dose
Amoxicillin (β-lactam)	Blocks cell-wall synthesis	Allergy, GI upset, *C. difficile*

Two more classic doxycycline facts come from standard pharmacology

- Antacids, dairy and iron reduce its absorption.
- Expired tetracyclines can cause Fanconi syndrome.

Follow-up and partners

- **Test of cure at 3 weeks** if the patient still has symptoms, is pregnant or got suboptimal treatment. This applies to both infections.
- **Retest everyone else at 3 months**, because reinfection is common.
- **Pharyngeal gonorrhea:** test of cure at 7–14 days.
- **Expedited partner therapy** means treating partners without examining them: cefixime 800 mg once for gonorrhea, and doxycycline 7 days or azithromycin 1 g once for chlamydia.

Source discrepancies:

- **Chlamydia in pregnancy:**
  - Some say amoxicillin alone.
  - Others = azithromycin with amoxicillin as the alternative (would go with this)
- **Covering chlamydia when treating gonorrhea:**
  - Treat "even if infection is ruled out."
  - Some say add it when chlamydia hasn't been ruled out.
- **Pharyngeal test-of-cure timing:** 7–14 days Or the general 3-week test of cure.

1. Discuss the potential long-term implications of HPV infection and the rationale for HPV vaccination. HSS (Please review general HPV information and HPV vaccine guidelines on the CDC website, with specific attention to cancers linked to HPV and vaccine recommendations. **Shannon**
   Available at: https://www.cdc.gov/hpv/index.html)

About HPV

- Human papillomavirus (HPV) - common virus that can cause cancers later in life.
  - Can protect from these cancers with HPV vaccination.
  - Nearly everyone who is not vaccinated will get HPV at some point in their lives.
- What is it
  - **HPV infections are very common.**
    - Nearly everyone will get HPV at some point in their lives.
  - HPV is spread through intimate skin-to-skin contact.
    - vaginal, anal, or oral sex with someone who has the virus, even if they don't have signs or symptoms.
- Types
  - Referred to as "non-oncogenic" (wart-causing) or "oncogenic" (cancer-causing)
    - Based on whether they put a person at risk for cancer.
- How it affects your body
  - Most HPV infections (9 out of 10) go away by themselves within 2 years.
  - Sometimes, HPV infections will last longer and can cause some cancers. *Discussed later*

Cancers Caused by HPV

- HPV can cause cancers of the:
  - Cervix, vagina, and vulva in women
    - More than 9 of every 10 cases of cervical cancer are caused by HPV. Almost all cervical cancer can be prevented by HPV vaccination.
  - Penis in men
    - About 4 out of every 10 cases of cancer caused by HPV occur among men.
  - Anus in both men and women
  - Back of the throat, including the base of the tongue and tonsils (oropharynx), in both men and women
- Cancer usually takes years, even decades, to develop after a person gets HPV.
  - **There is no way to know who will develop cancer or other health problems from HPV.**
- People with weakened immune systems (like people living with HIV/AIDS) may be less able to fight off (clear) HPV infections, and more likely to develop health problems
- Cervical cancer = Once the leading cause of cancer deaths among women in the US
  - HPV vaccines and cervical cancer screening have made it one of the most preventable cancers.
  - Even women who are vaccinated should still get regular cervical cancer screening starting at age 21.
  - Unique: can be detected early by a recommended screening test
    - Other types of cancer caused by HPV may not be found until they cause more serious health problems.

Prevention/Vaccination

- **Protect with vaccination**; provides safe, effective, and lasting protection against the HPV infections that most commonly cause cancer
    - Fewer teens and young adults are getting genital warts.
    - HPV vaccination has also reduced the number of cervical precancer cases in young women.
- HPV vaccination can prevent over 90% of cancers caused by HPV.
- **Children ages 11–12 years** should get 2 doses of HPV vaccine, given 6 to 12 months apart. HPV vaccines can be given starting at age 9.
    - 1st dose: 11–12 years old (can start at age 9)
    - 2nd dose: 6–12 months after the 1st dose
    - Rationale: The HPV vaccine series is most effective when given before a person is exposed to the virus.
        - Give at age 12-11 = before contact with the HPV virus
- Children who get the first dose **before** their 15th birthday need only 2 doses.
- **Children 9–14 years old** who have received 2 doses of HPV vaccine less than 5 months apart will need a third dose.
- Teens who get the first dose **on or after** their 15th birthday need 3 doses.
- **People 15–26 years old** who start the series later need 3 doses of HPV vaccine.
    - The doses are given over 6 months.
- **People with weakened immune systems** should get 3 doses if they are 9–26 years old.
- **People older than 26 years.** Vaccination is not recommended for everyone older than age 26 years.
    - Some adults age 27 through 45 years who are not already vaccinated may decide to get HPV vaccine after speaking with their doctor about their risk for new HPV infections and the possible benefits of vaccination for them.
    - HPV vaccination in this age range provides less benefit, because more people in this age range have already been exposed to HPV.
- Types of vaccines:
    - Gardasil-9 (9vHPV) is the vaccine distributed in the US.
        - Protects against nine HPV types (6, 11, 16, 18, 31, 33, 45, 52, and 58).
    - The quadrivalent HPV vaccine (Gardasil, 4vHPV) and bivalent HPV vaccine (Cervarix, 2vHPV) were licensed by the U.S. FDA
- All of the HPV vaccines protect against HPV types **16 and 18** that cause most HPV cancers.
- Who should get vaccinated
    - See above guidelines
- Who should NOT get vaccinated
    - Life-threatening allergic reaction to any ingredient of an HPV vaccine, or to a previous dose of HPV vaccine.
    - Have an allergy to yeast (Gardasil and Gardasil 9).
    - Are pregnant.
- HPV vaccines are safe for children who are mildly ill.
    - Includes those with a low-grade fever of less than 101 degrees, a cold, runny nose, or cough.
- People with a moderate or severe illness → wait until they are better.
- The protection provided by HPV vaccines lasts a long time.

    - People who received HPV vaccines were followed for at least about 12 years, and their protection against HPV has remained high with no evidence of decreasing over time.
- Side effects:
    - Many people → no side effects at all; but = mot common side effects of HPV vaccine are usually mild and include:
        - Pain, redness, or swelling in the arm where the shot was given
        - Fever
        - Dizziness or fainting (fainting after any vaccine, including HPV vaccine, is more common among adolescents than others)
        - Headache or feeling tired
        - Nausea
        - Muscle or joint pain
    - To prevent fainting and injuries from fainting, teens should be seated or lying down during vaccination; they should also do the same for 15 minutes after getting the shot.
- Finding the vaccine
    - doctor's office = the best place to receive recommended vaccines.
        - Annual checkups
        - Sports physicals
        - Annual flu shot or other vaccine visits
    - If your doctor does not stock HPV vaccine, ask for a referral. *MAJOR SHADE but true lol*
    - Vaccines = available at pharmacies, workplaces, community health clinics, health departments, or schools.
- Paying for the vaccine
    - Most health insurance plans cover vaccine costs.
    - Children Program = may be able to get no-cost vaccines through the Vaccines for Children (VFC) Program.
        - Helps families of eligible children who may not be able to afford or have access to vaccines.